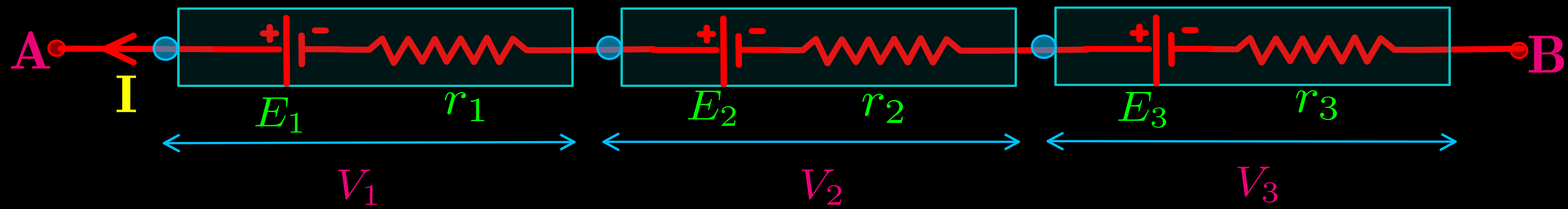


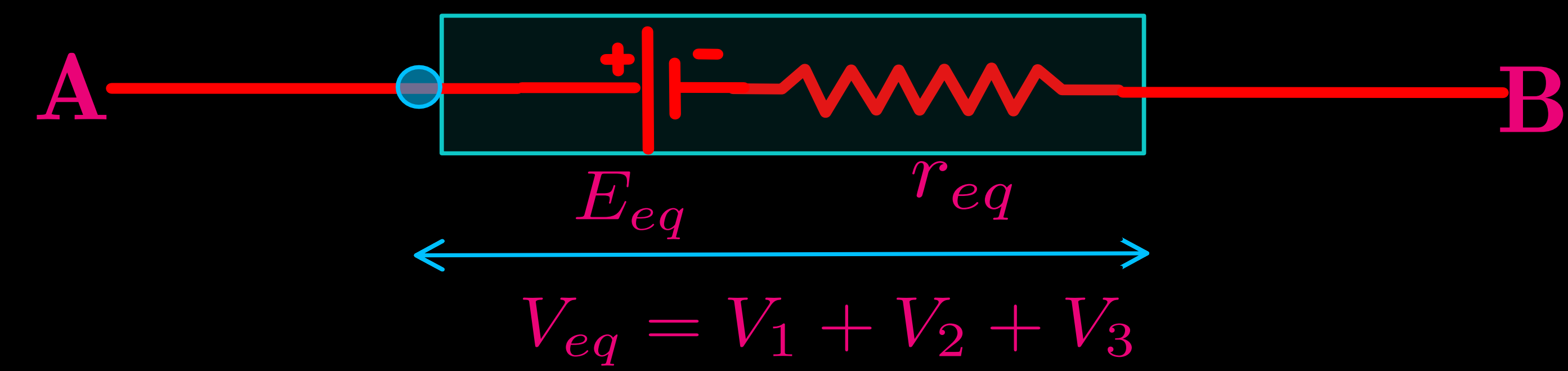
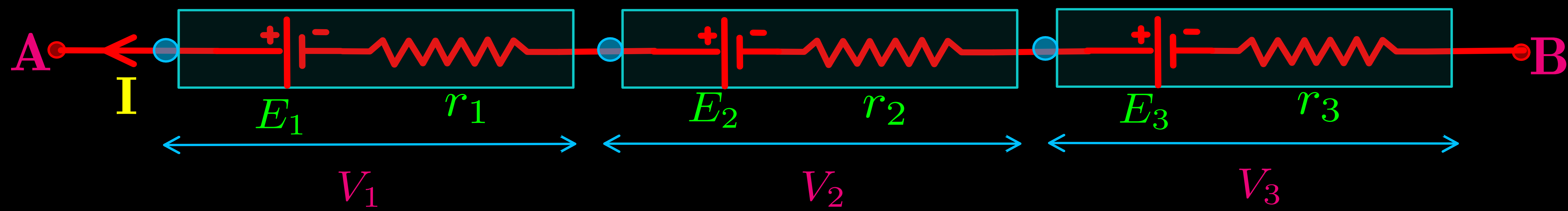
CELLS IN SERIES

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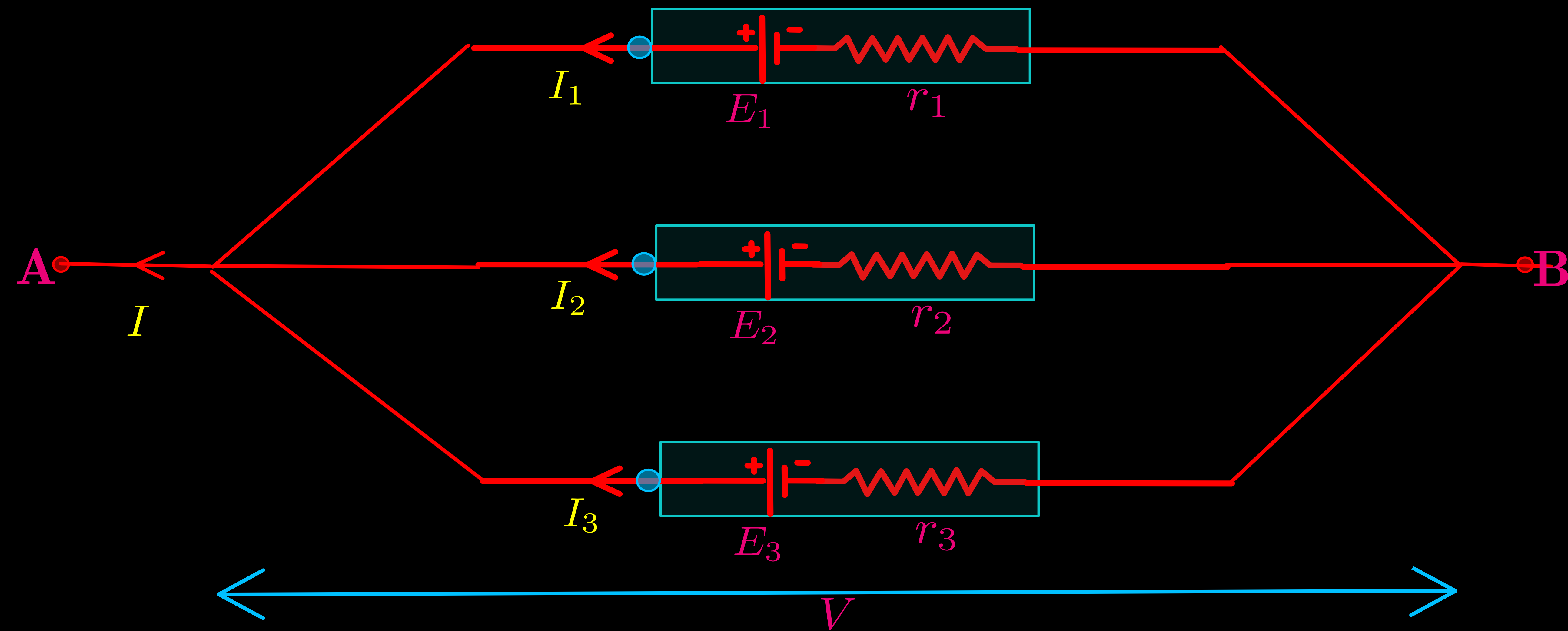
CELLS IN SERIES



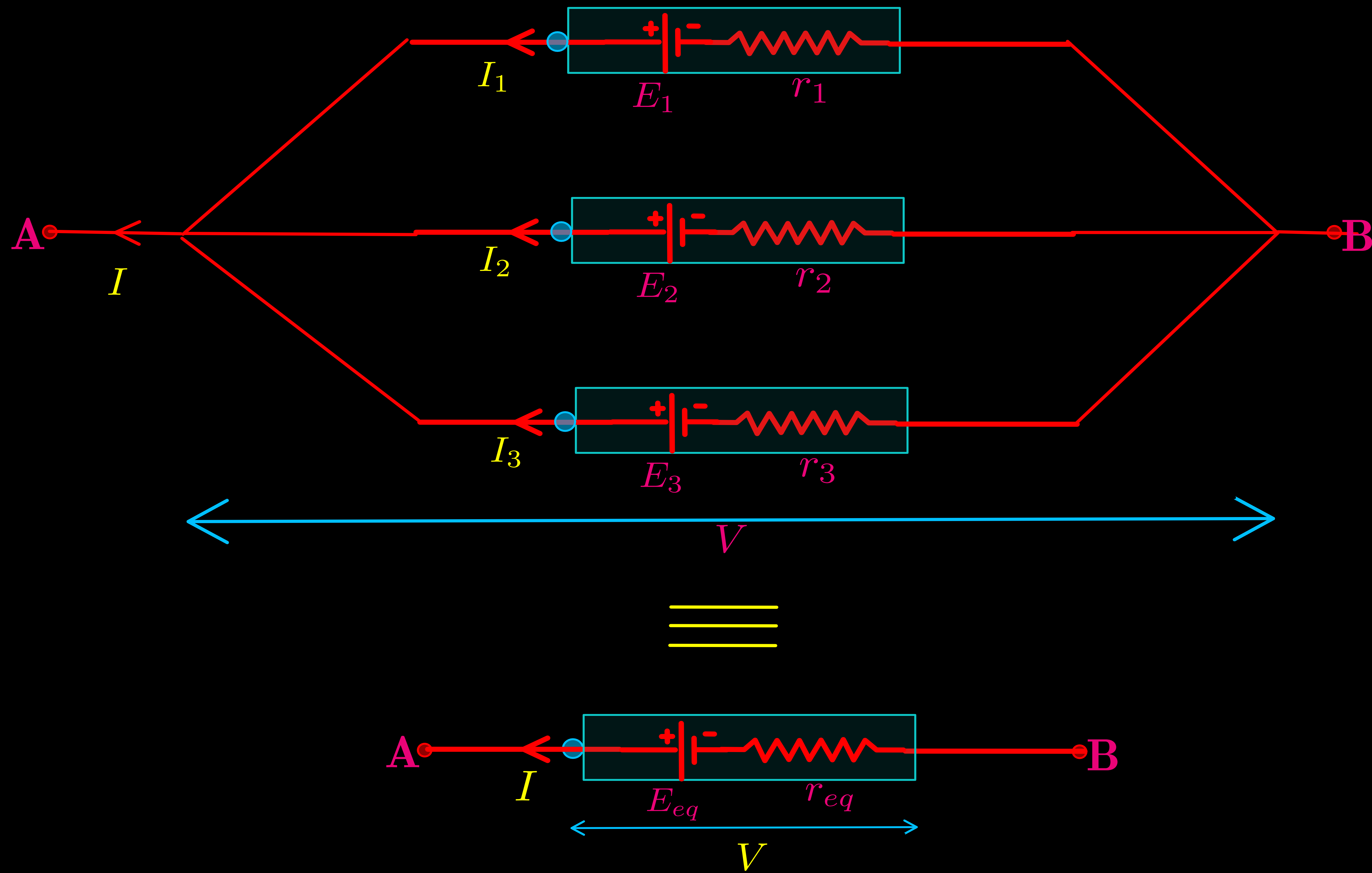
CELLS IN SERIES



CELLS IN PARALLEL



CELLS IN PARALLEL



KIRCHHOFF'S RULES

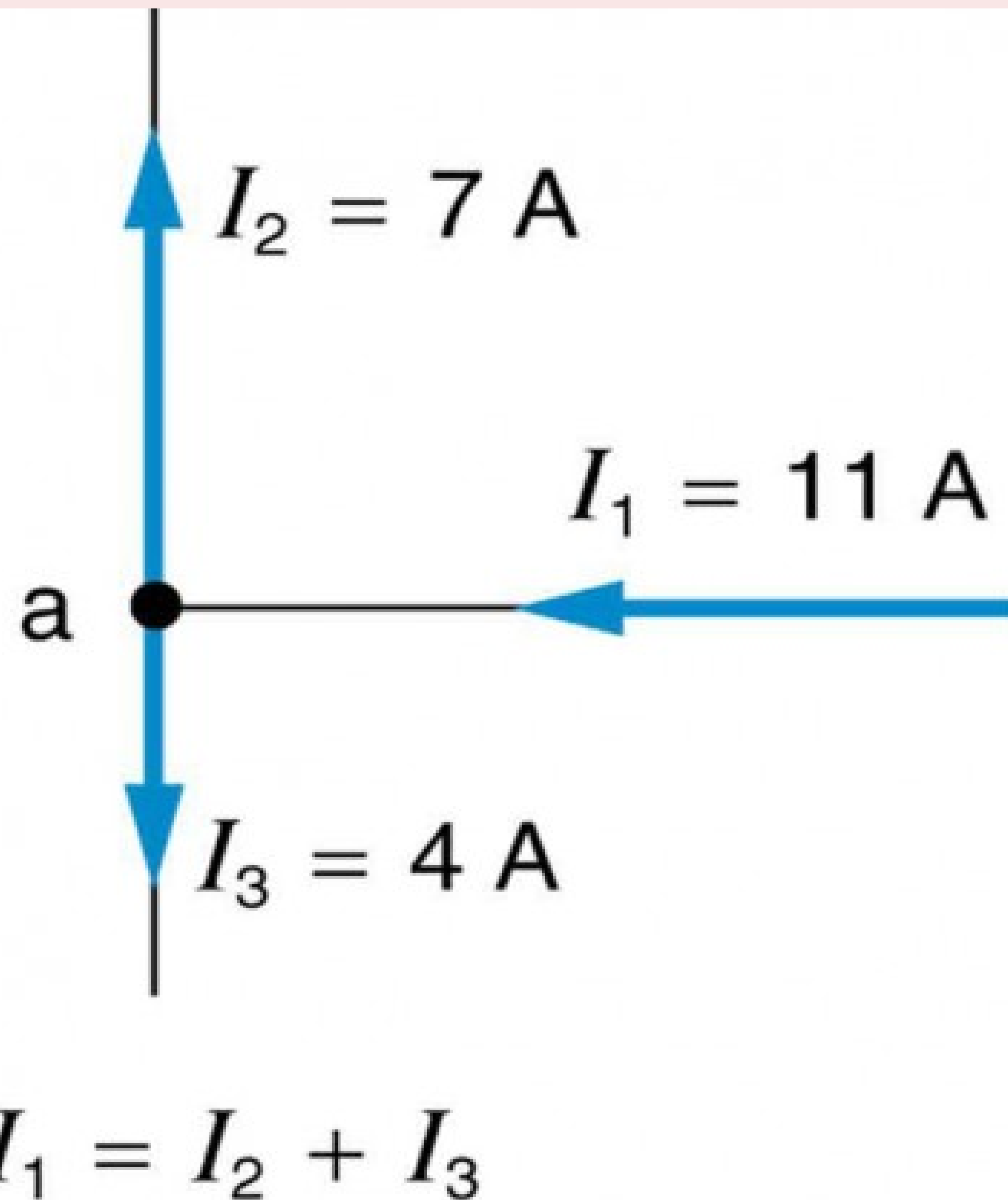
Chapter [3] - Current Electricity
Class - XII

Manish Joshi

D.A.V. Lohaghat

August 22, 2024

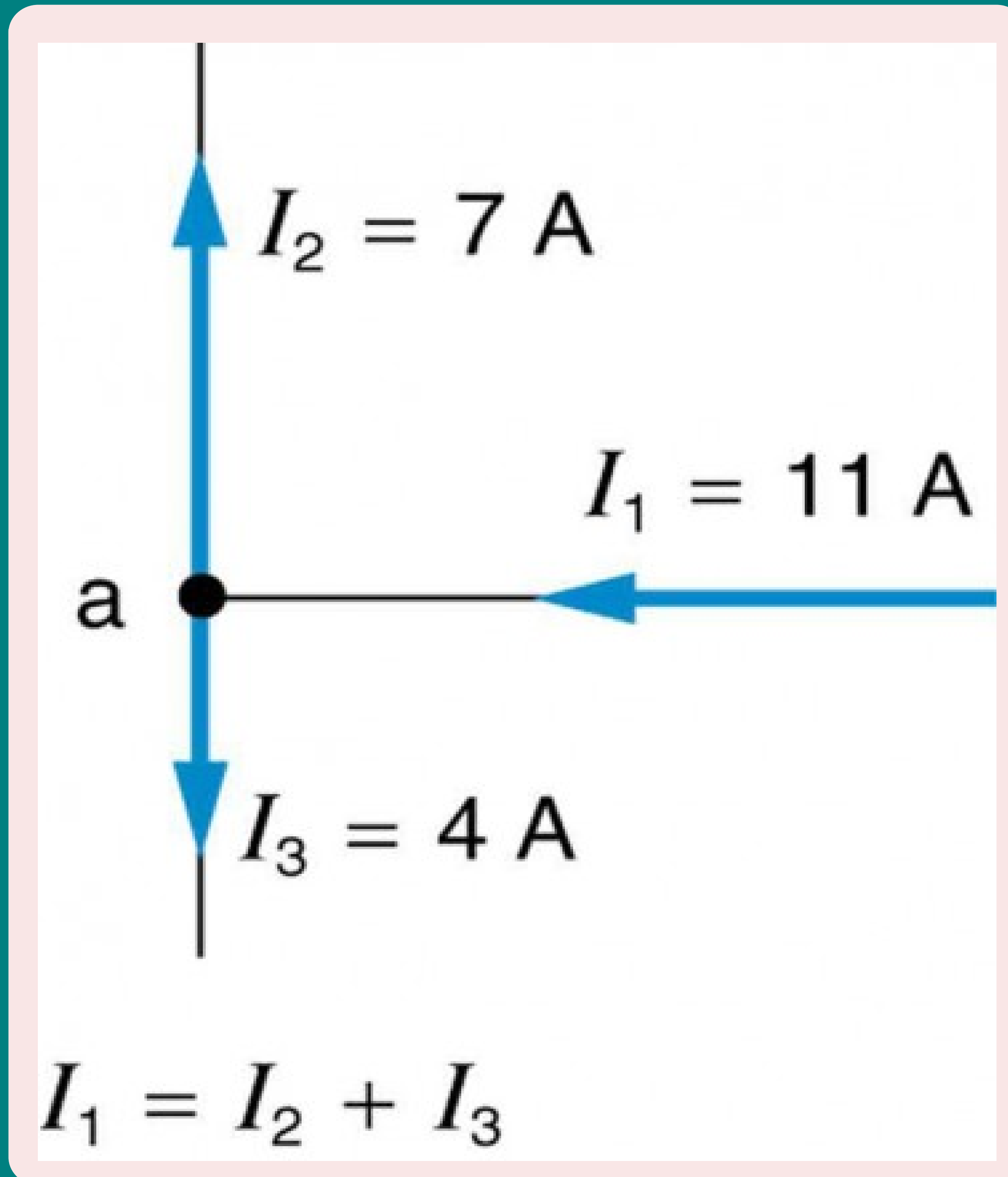
KIRCHHOFF'S RULES



Junction rule:

- At any **junction**, the **sum of the currents entering** the junction is **equal** to the **sum of currents leaving** the junction.

KIRCHHOFF'S RULES

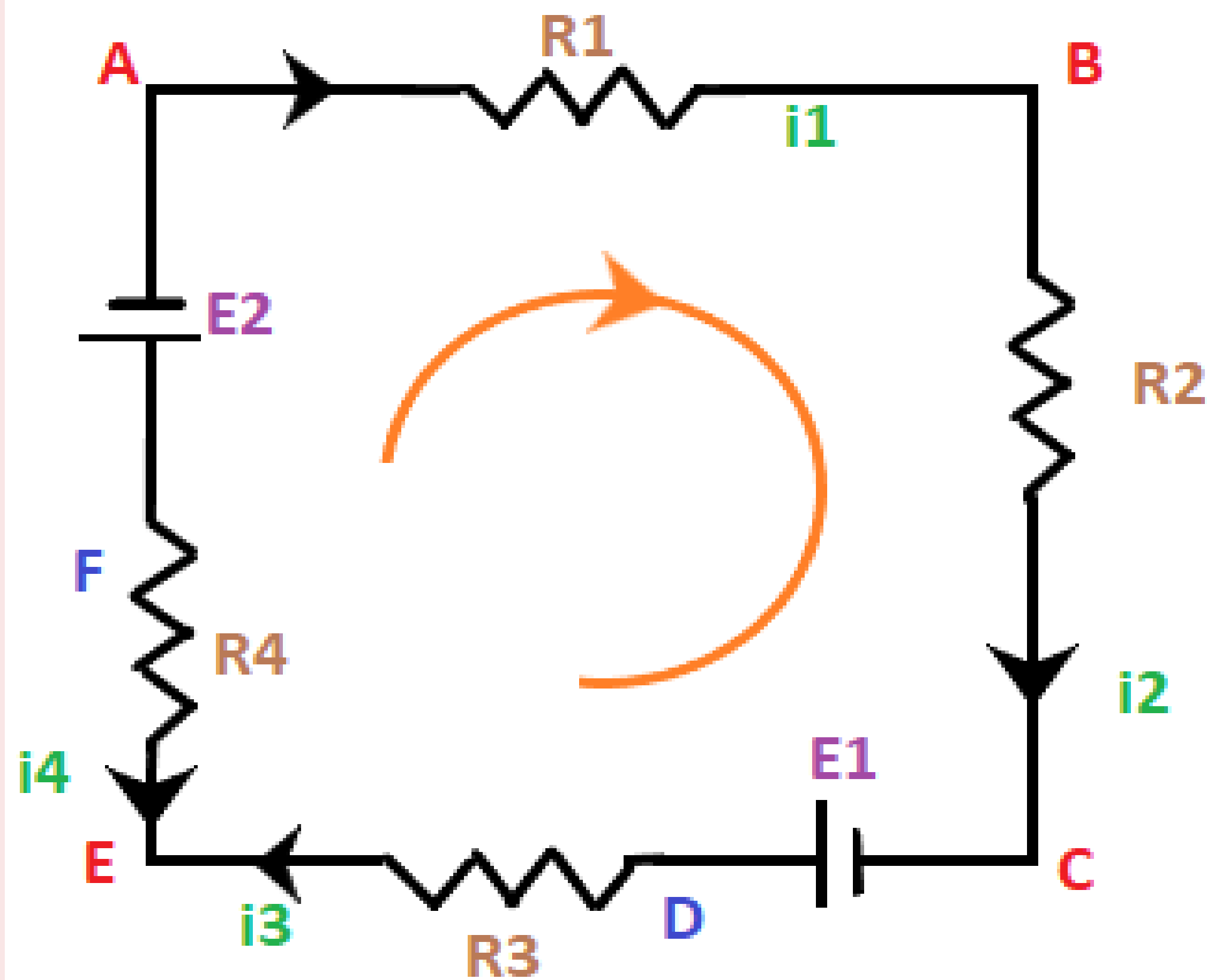


Junction rule:

- At any **junction**, the **sum of the currents entering** the junction is **equal** to the **sum of currents leaving** the junction. This is a consequence of **charge conservation**.

$$\sum I = 0$$

KIRCHHOFF'S RULES



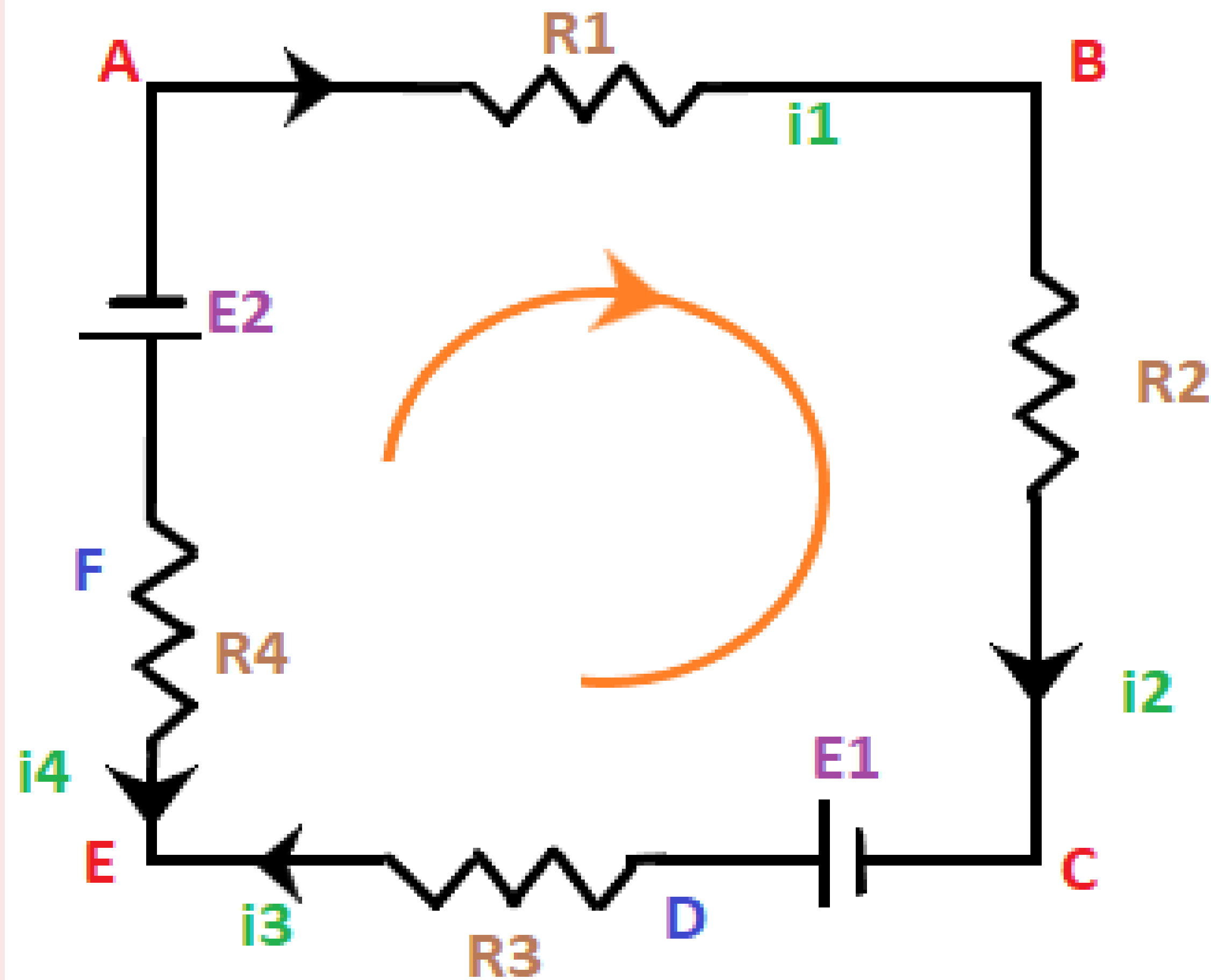
$$E_1 - E_2 = i_3 R_3 - i_4 R_4 + i_1 R_1 + i_2 R_2$$

Loop rule:

- It states that around each **loop** in an electric circuit the **sum of the emf's is equal** to the **sum of the potential drops, or voltages across each of the resistances**, in the same loop.

$$\sum \xi = \sum IR$$

KIRCHHOFF'S RULES



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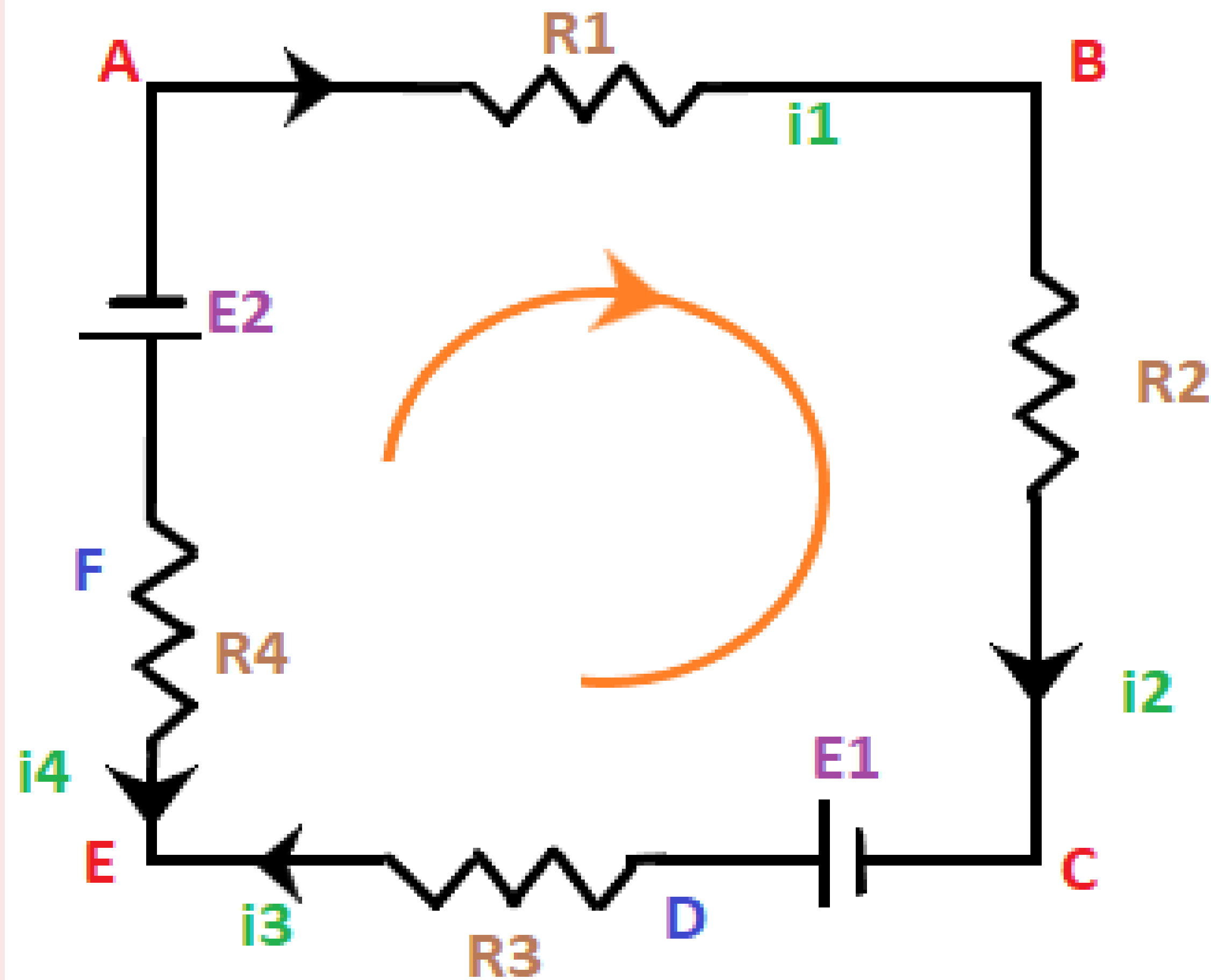
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KIRCHHOFF'S RULES



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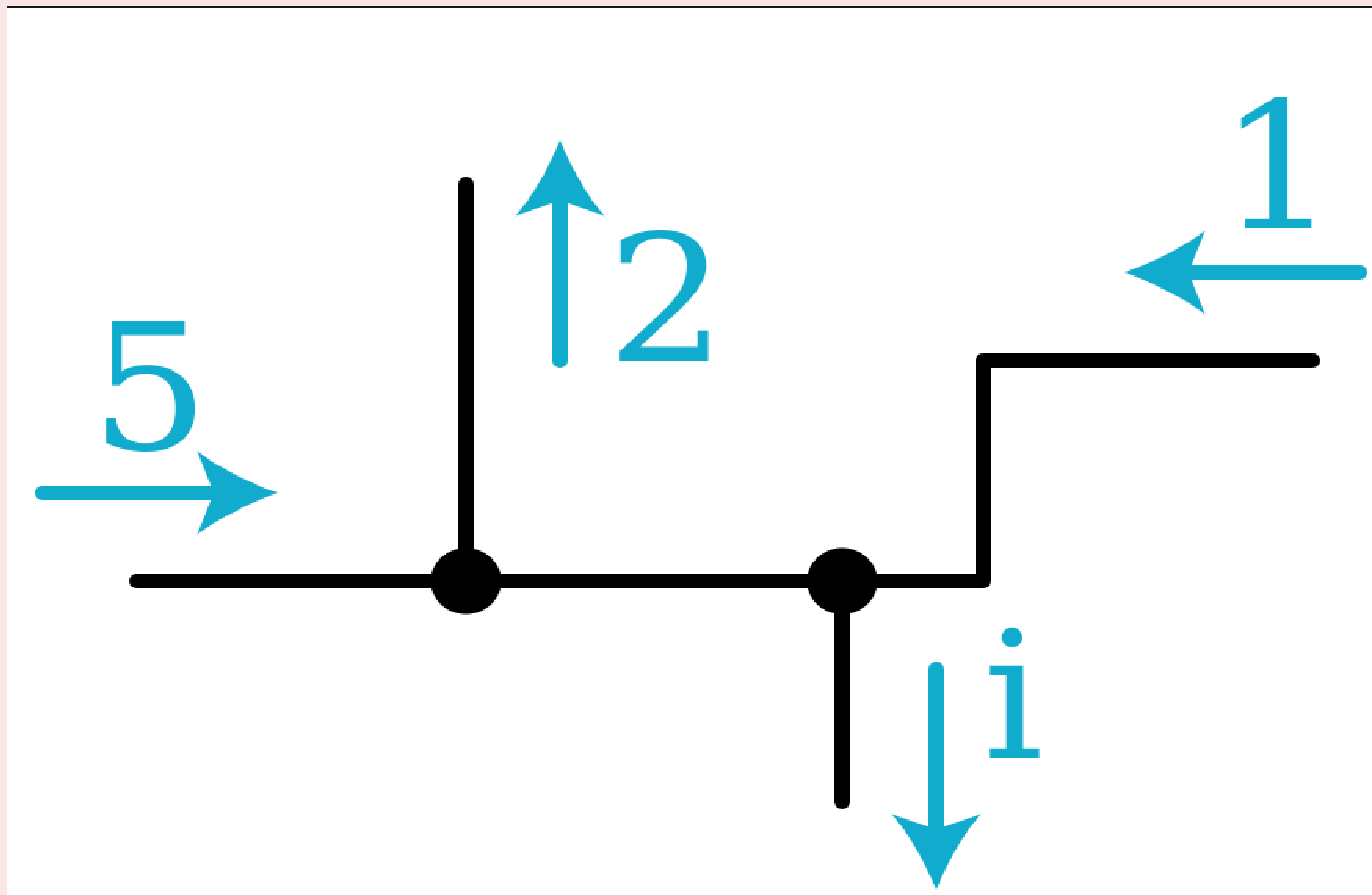
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- The **loop rule** is an application of **conservation of energy**. In a closed loop, whatever energy is supplied by emf must be transferred into other forms by devices in the loop.

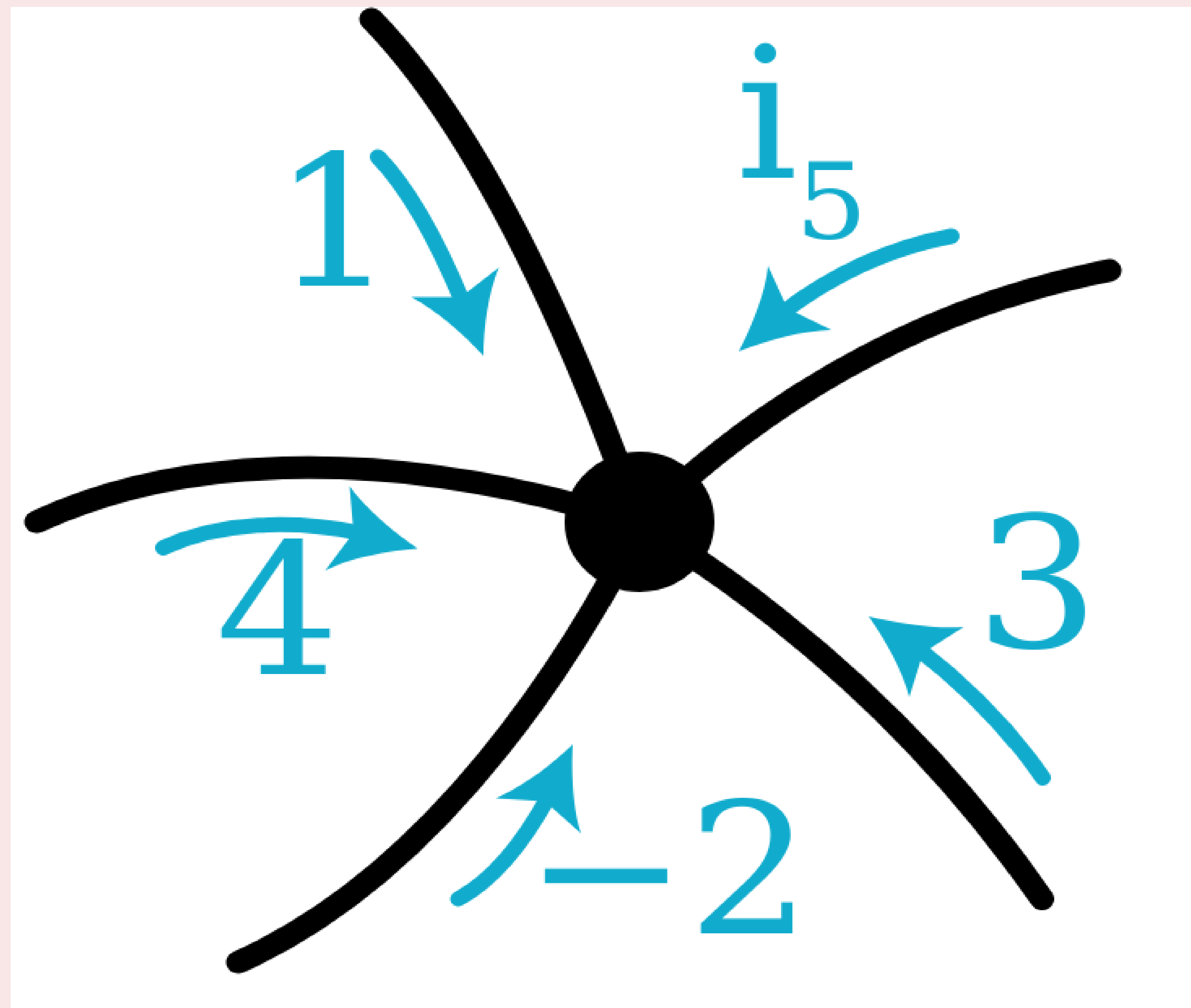
Kirchhoff's Current Law - concept checks

Problem 1: What is i ?



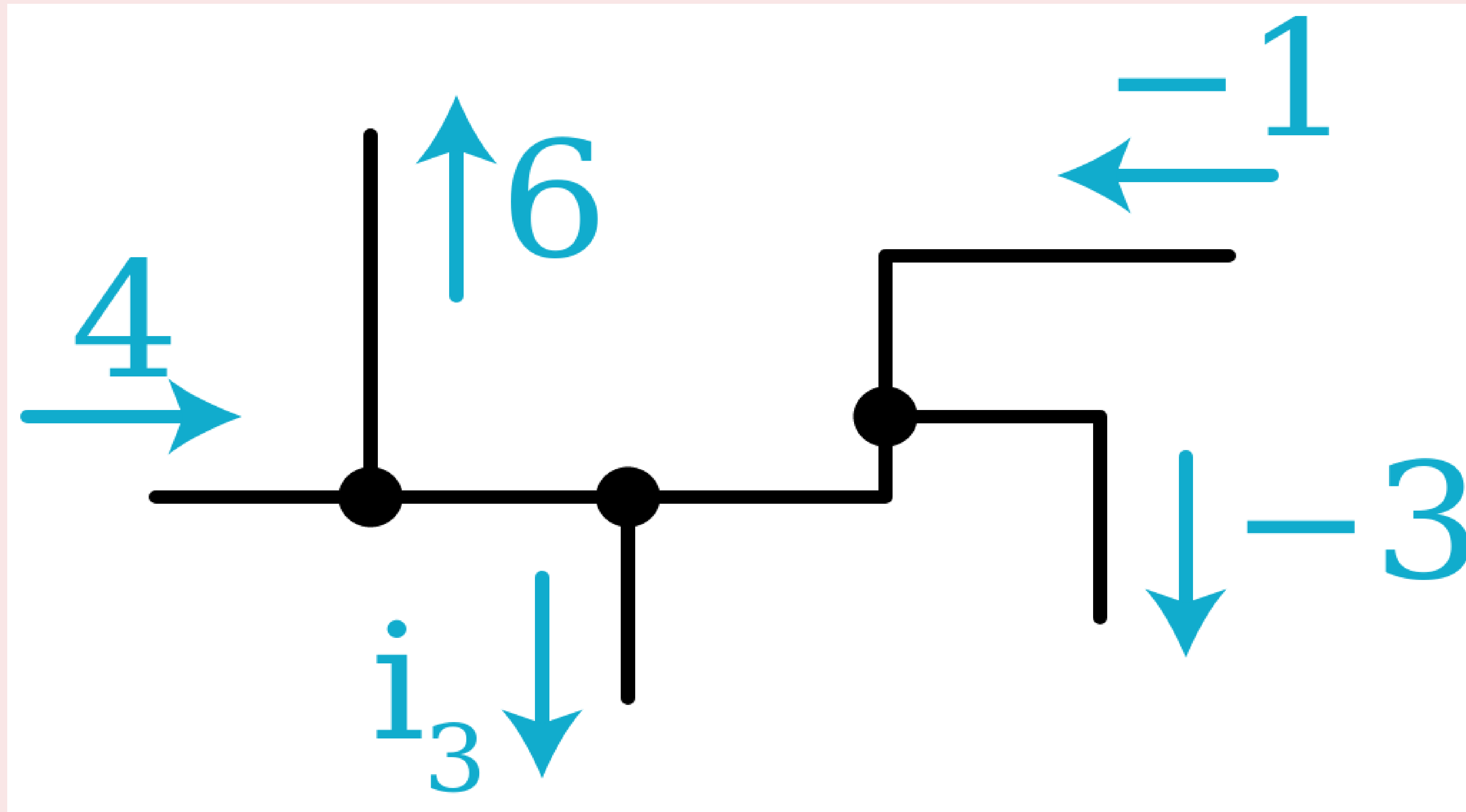
Kirchhoff's Current Law - concept checks

Problem 2 : What is i_5 ?



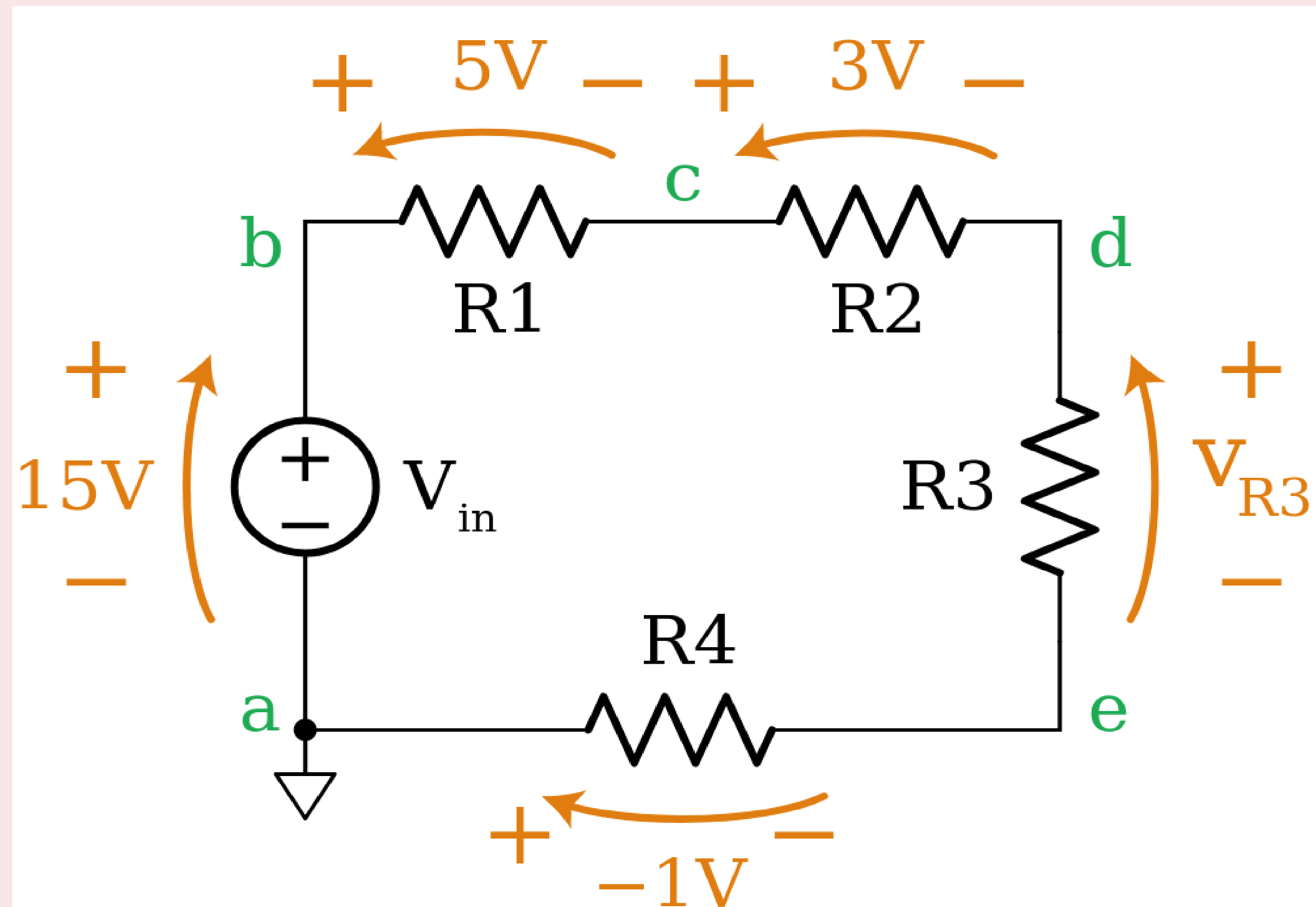
Kirchhoff's Current Law - concept checks

Problem 3: What is i_3 ?



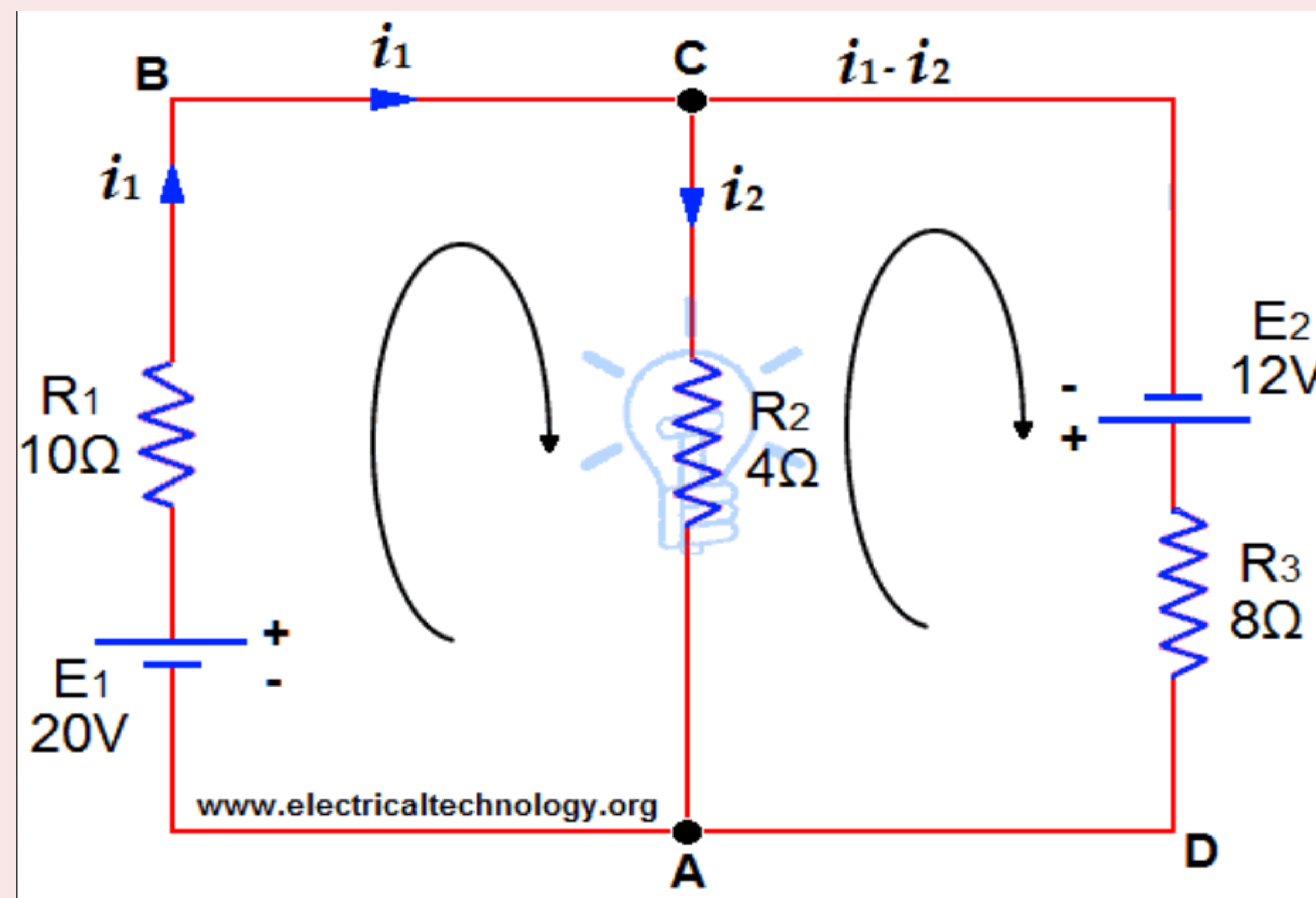
Kirchhoff's Voltage Law - concept check

Problem 4: What is V_{R3} ?

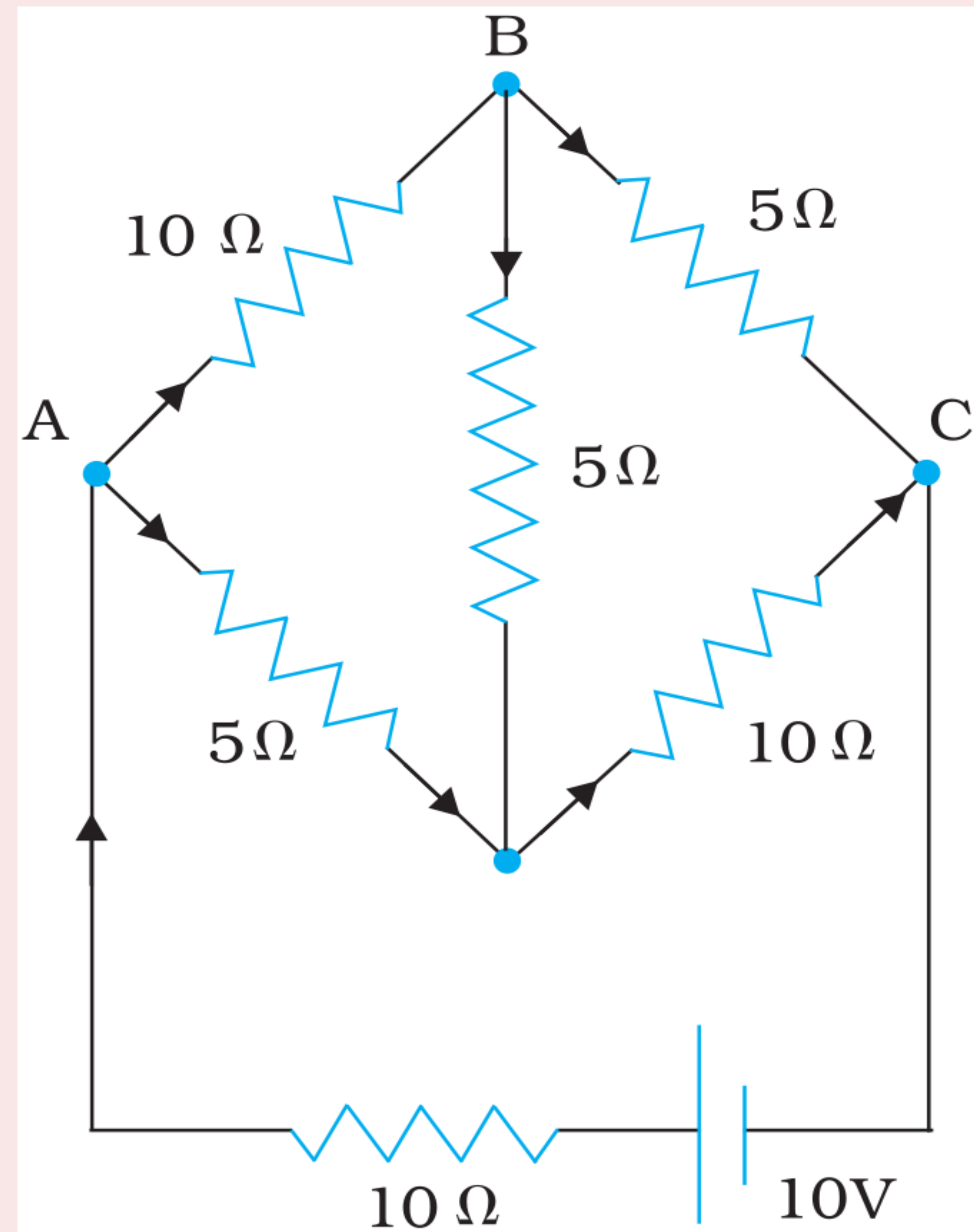


Kirchhoff's Voltage Law - concept check

Problem 5: Resistors of $R_1 = 10\Omega$, $R_2 = 4\Omega$ and $R_3 = 8\Omega$ are connected up to two batteries as shown. Find the current through each resistor.



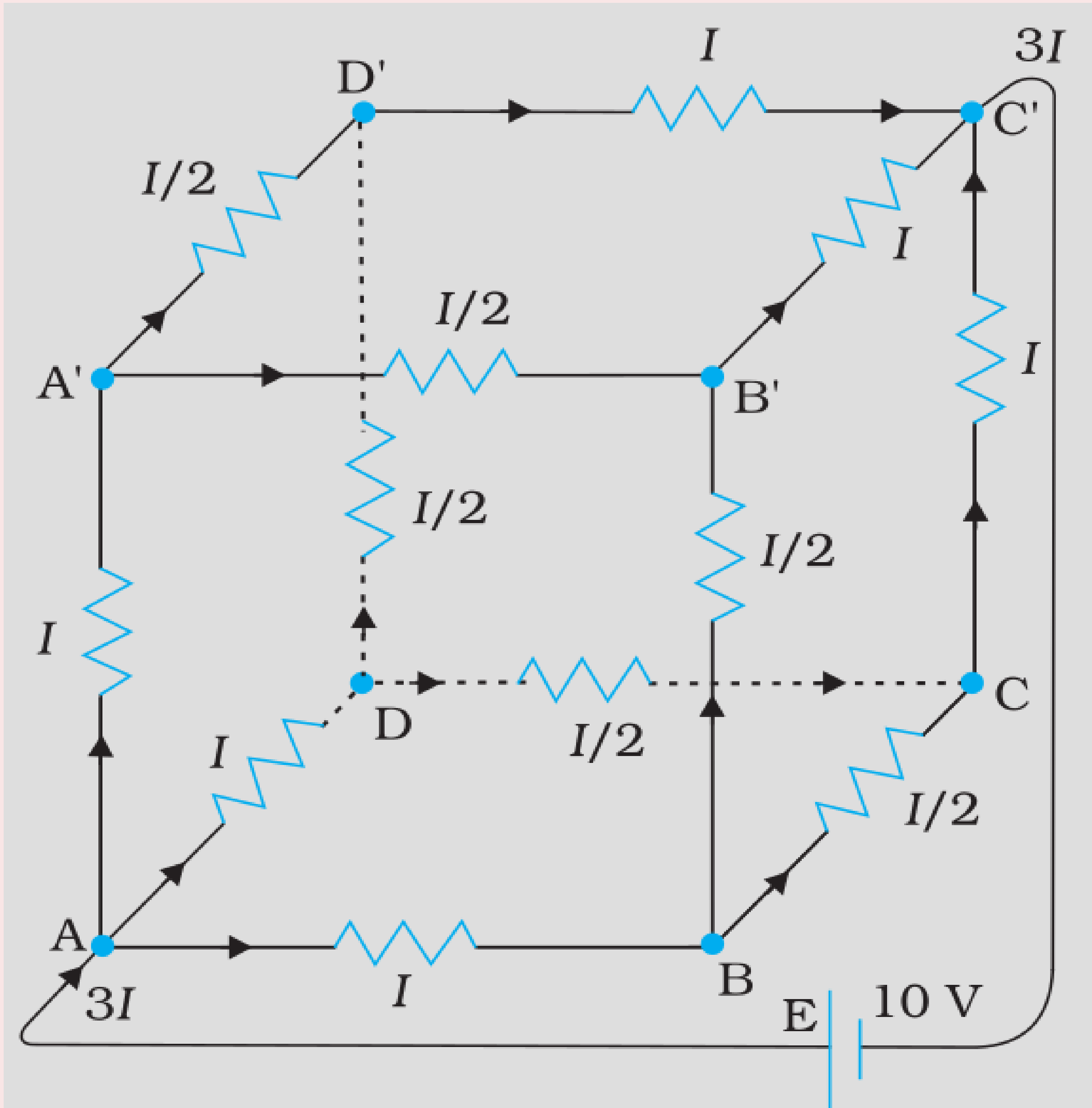
Kirchhoff's Laws



Problem 6:

Determine the current in each branch of the network shown in Fig.

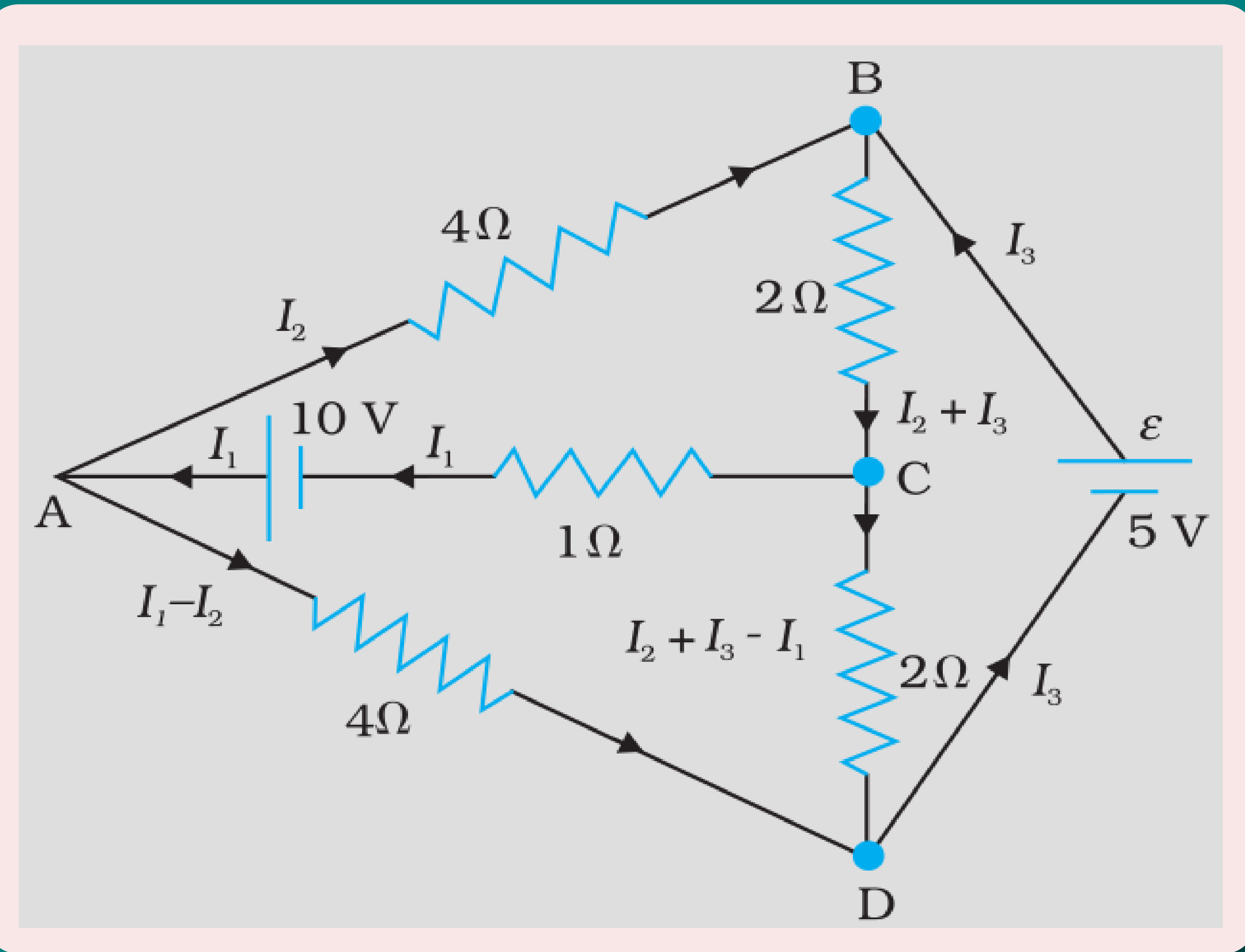
Kirchhoff's Laws



Problem 7:

A battery of 10 V and negligible internal resistance is connected across the diagonally opposite corners of a cubical network consisting of 12 resistors each of resistance $1\ \Omega$ (Fig. 3.23). Determine the equivalent resistance of the network and the current along each edge of the cube.

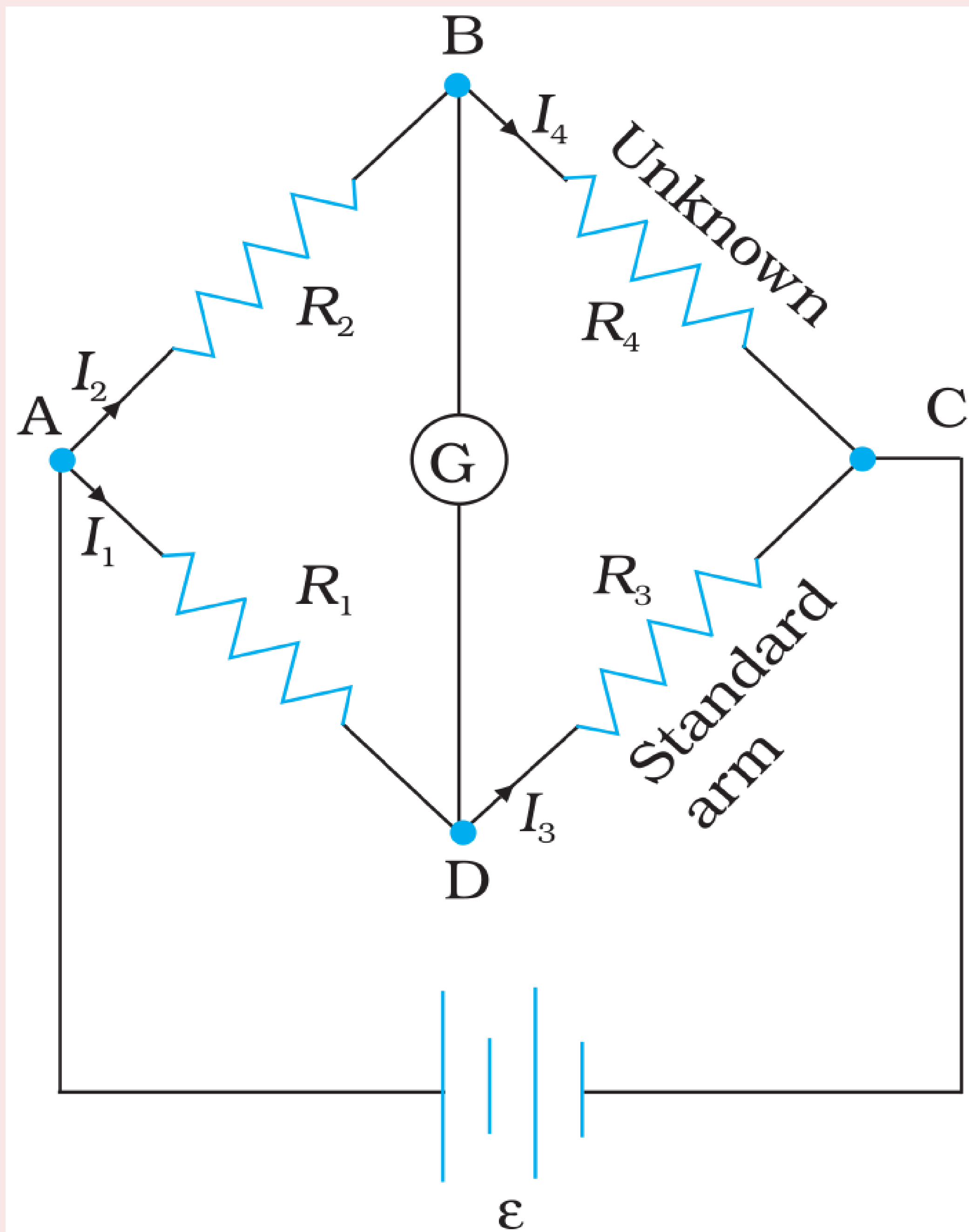
Kirchhoff's Laws



Problem 8:

Determine the current in each branch of the network shown in Fig.

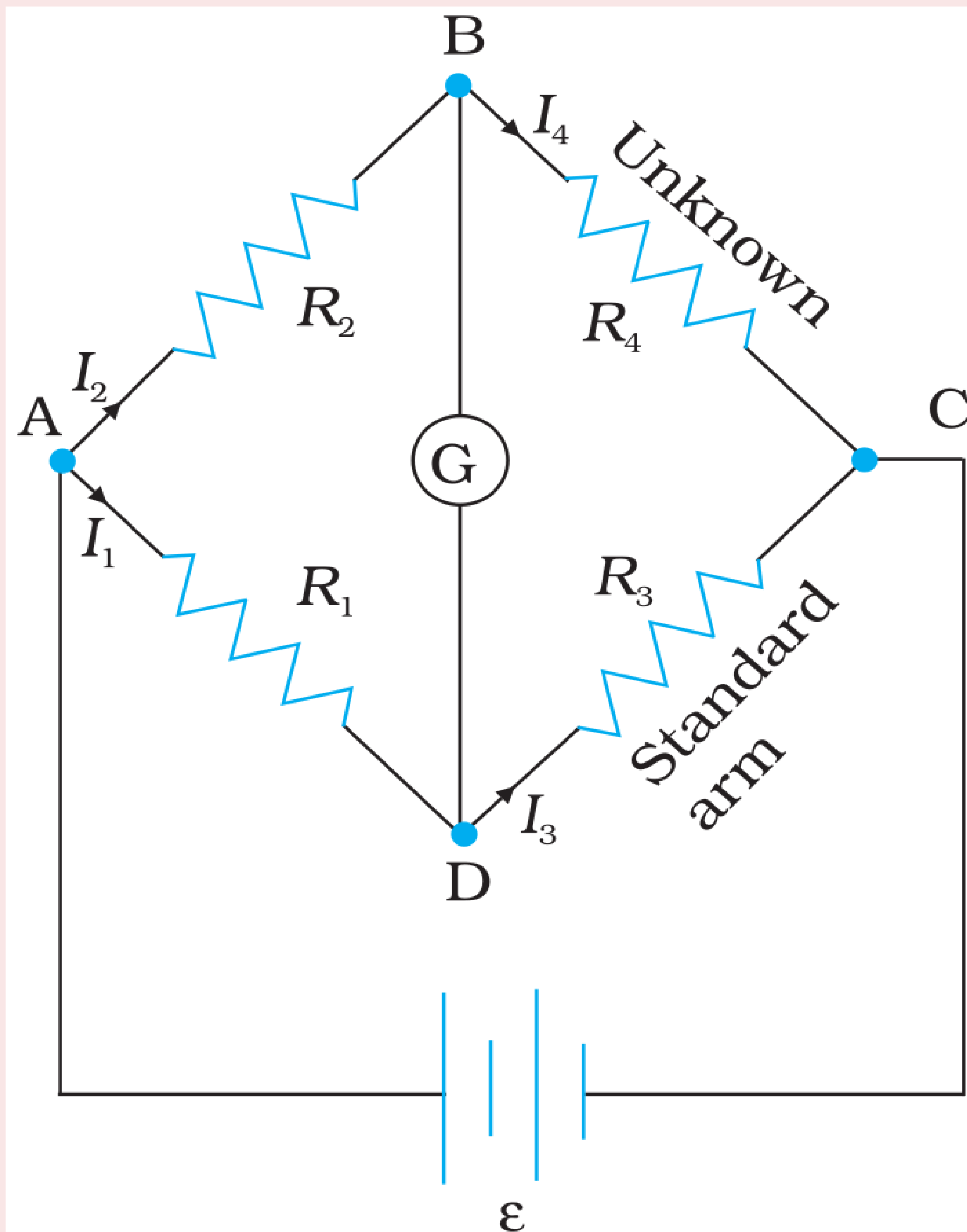
WHEATSTONE BRIDGE



Construction of Wheatstone Bridge:

- The **Wheatstone bridge** has four resistors R_1 , R_2 , R_3 and R_4 .

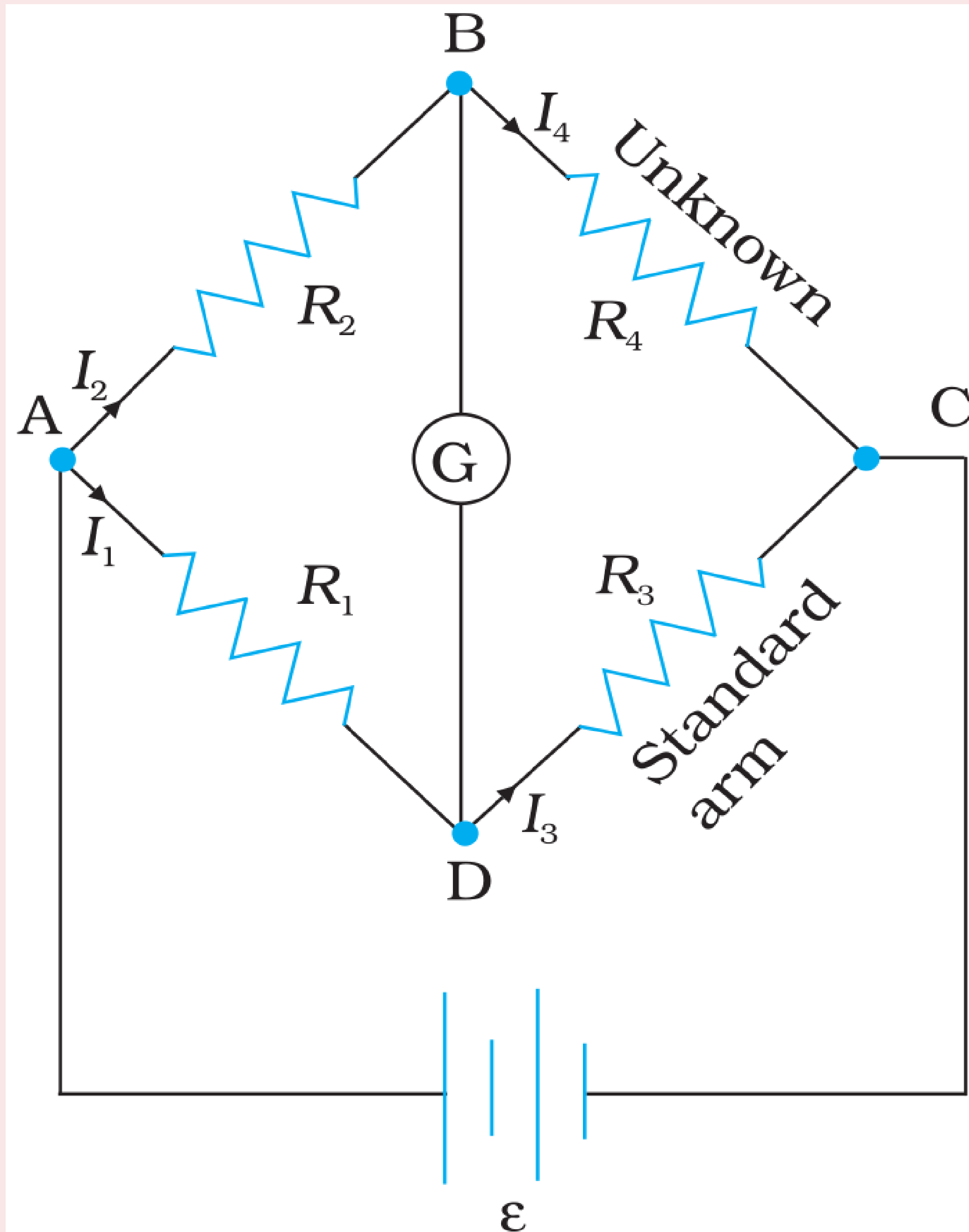
WHEATSTONE BRIDGE



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- Across one pair of diagonally opposite points (**A** and **C** in the figure) a **source (battery)** is connected.

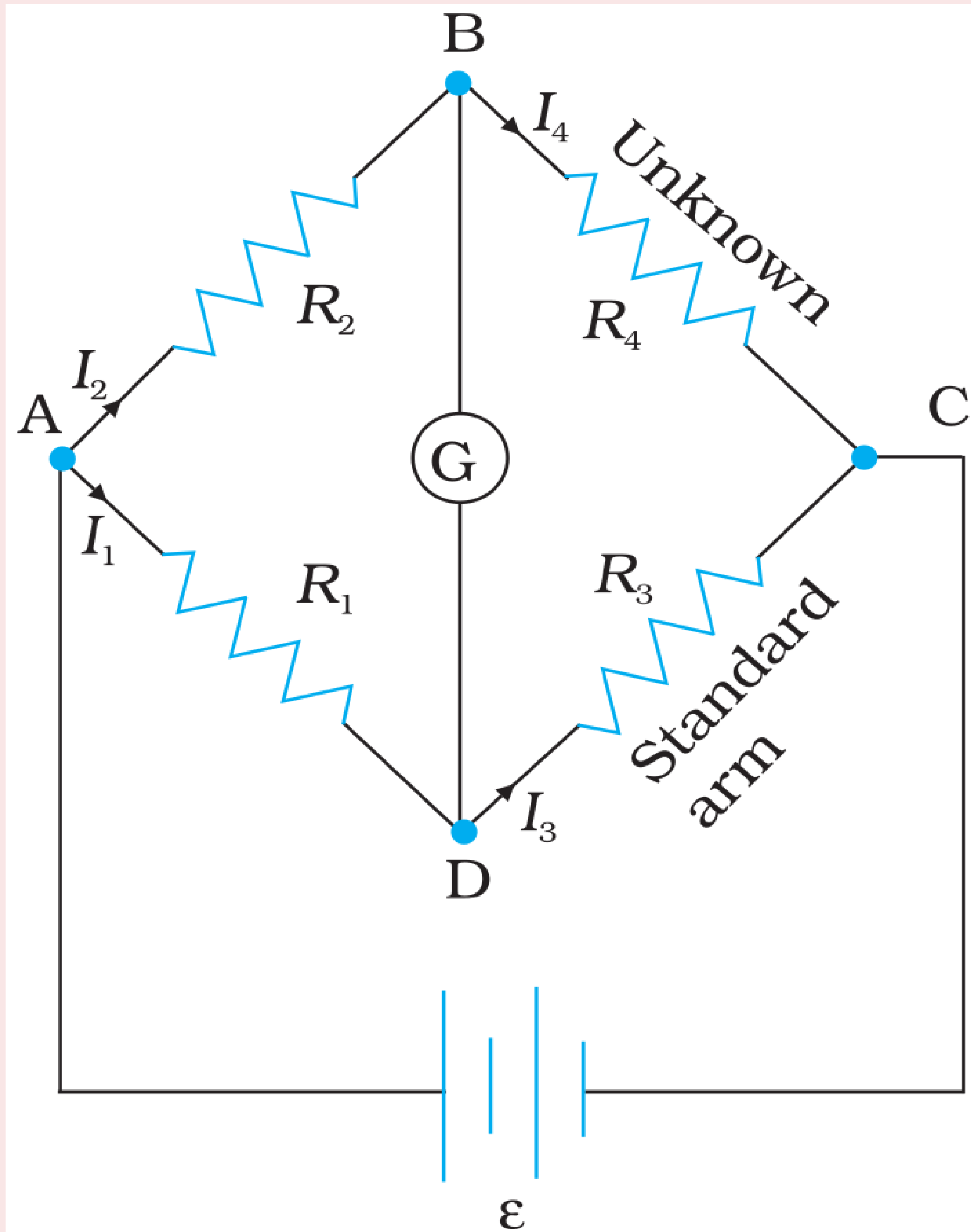
WHEATSTONE BRIDGE



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- The **Wheatstone bridge** has four resistors R_1 , R_2 , R_3 and R_4 .
- Across one pair of diagonally opposite points (**A** and **C** in the figure) a **source (battery)** is **connected**.
- Between the other two vertices, **B** and **D**, a **galvanometer G** (which is a **device to detect currents**) is **connected**.

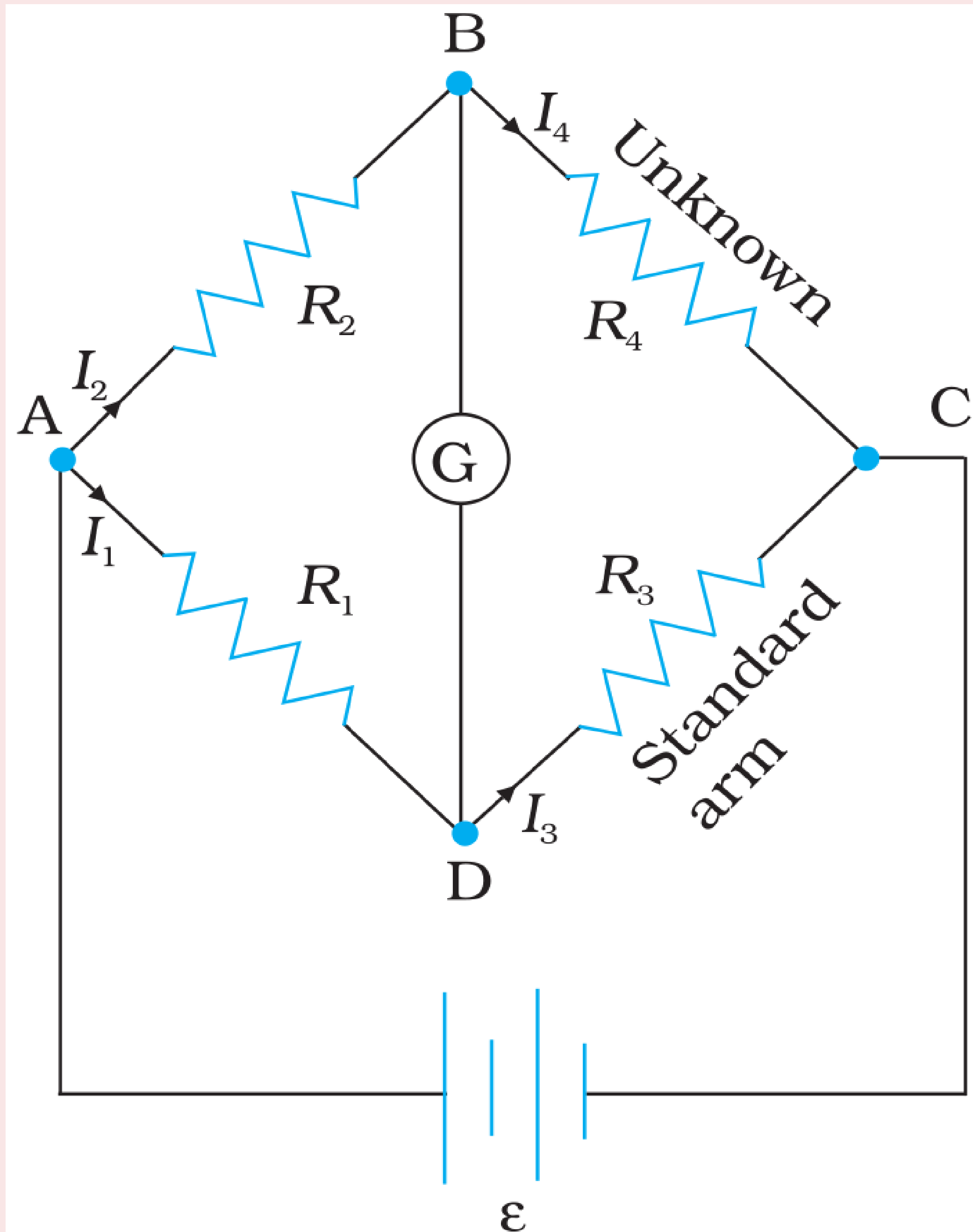
WHEATSTONE BRIDGE



Derivation: Balanced Bridge

- In general there will be currents flowing across all the resistors as well as a current I_g through G .

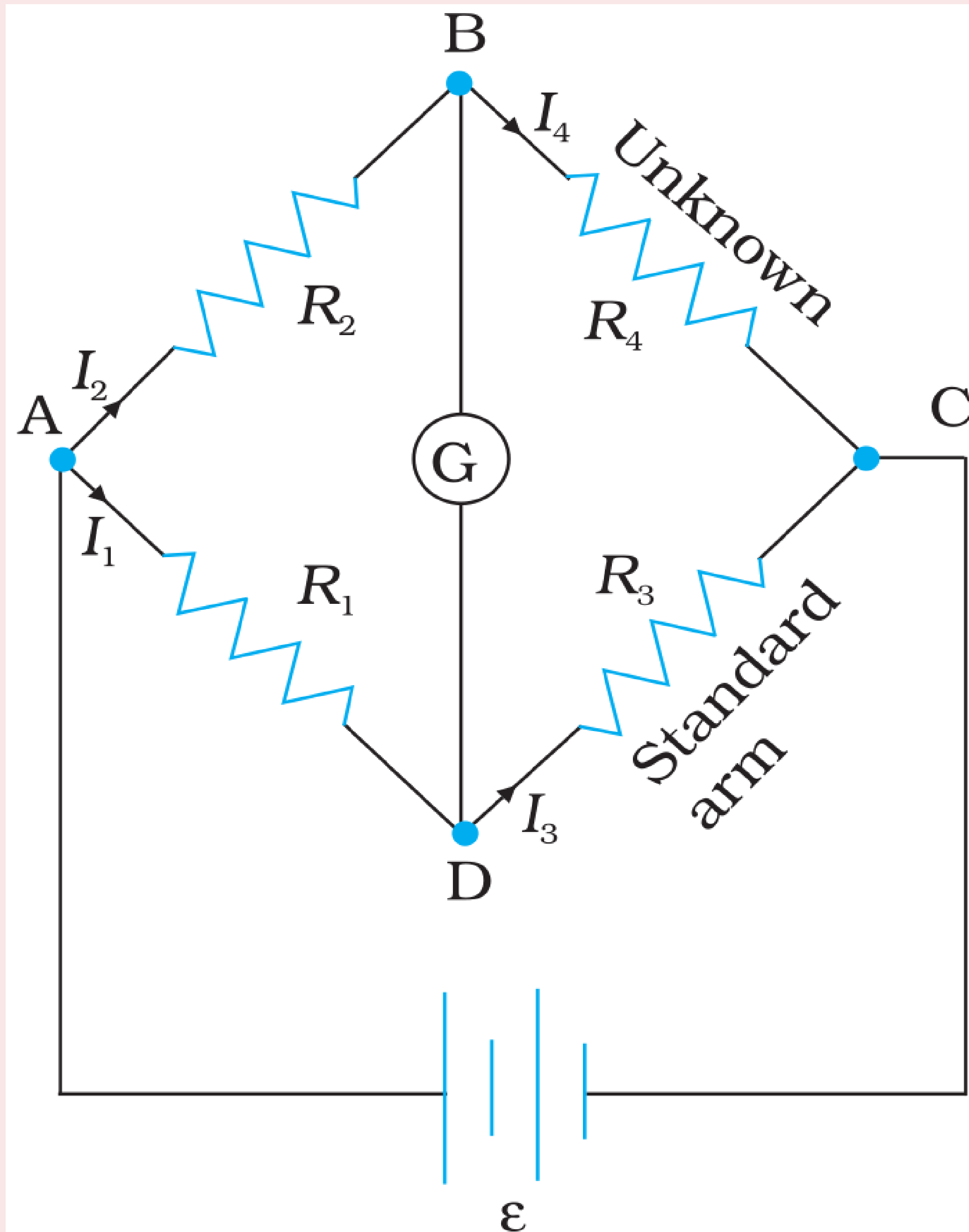
WHEATSTONE BRIDGE



Derivation: Balanced Bridge

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- In the case of a **balanced bridge** the resistors are such that there is **no current through G** (or $I_g = 0$).

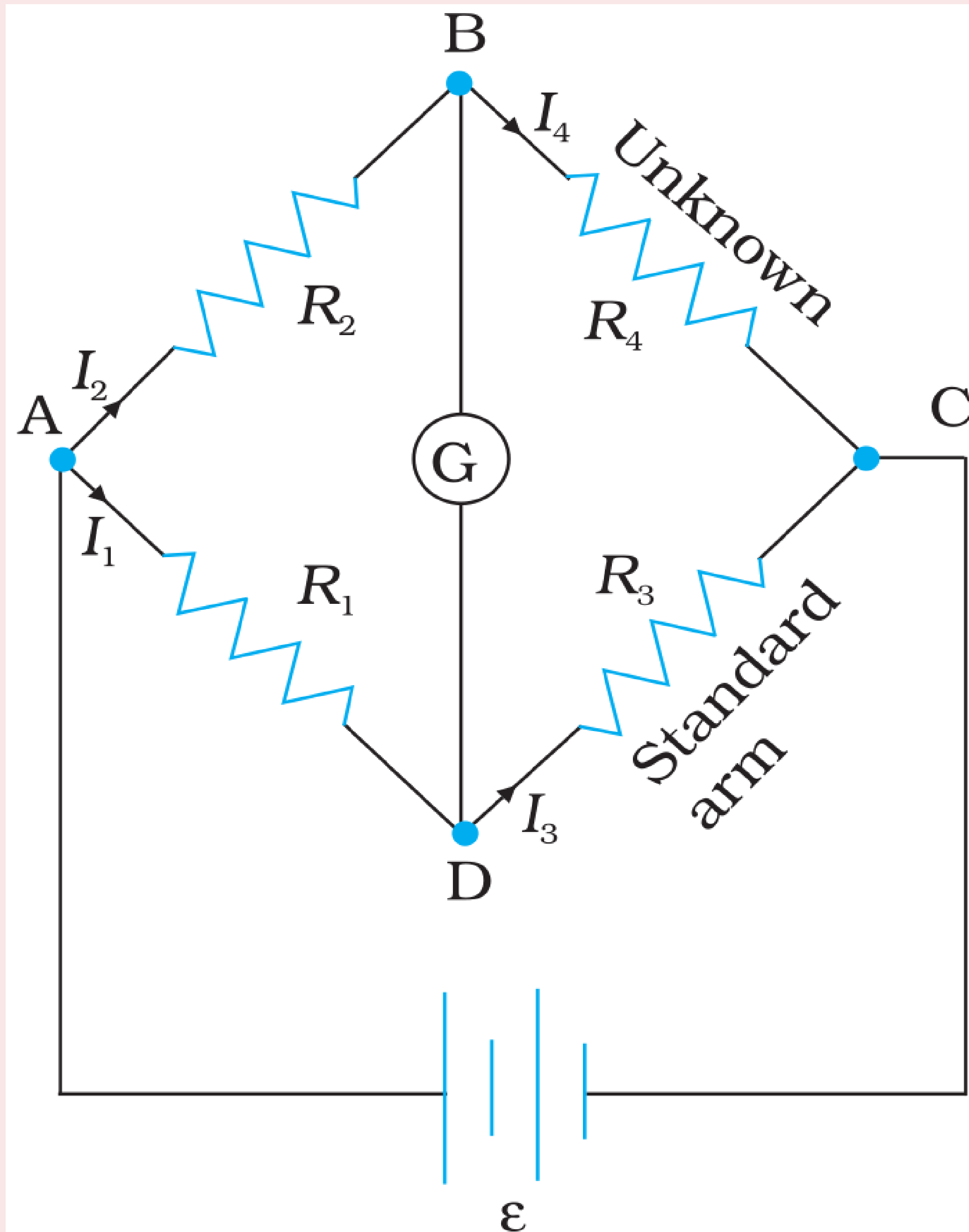
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WHEATSTONE BRIDGE

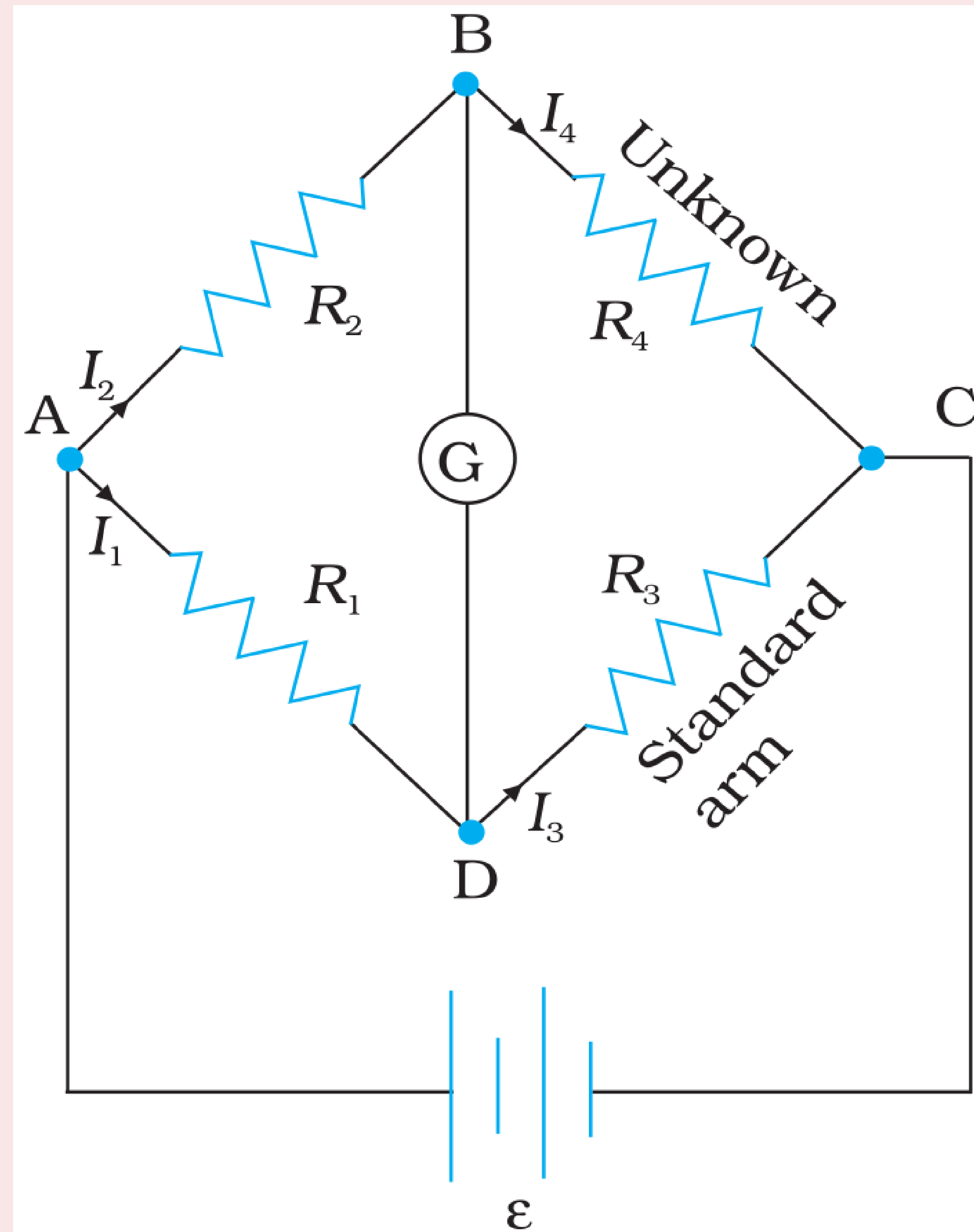


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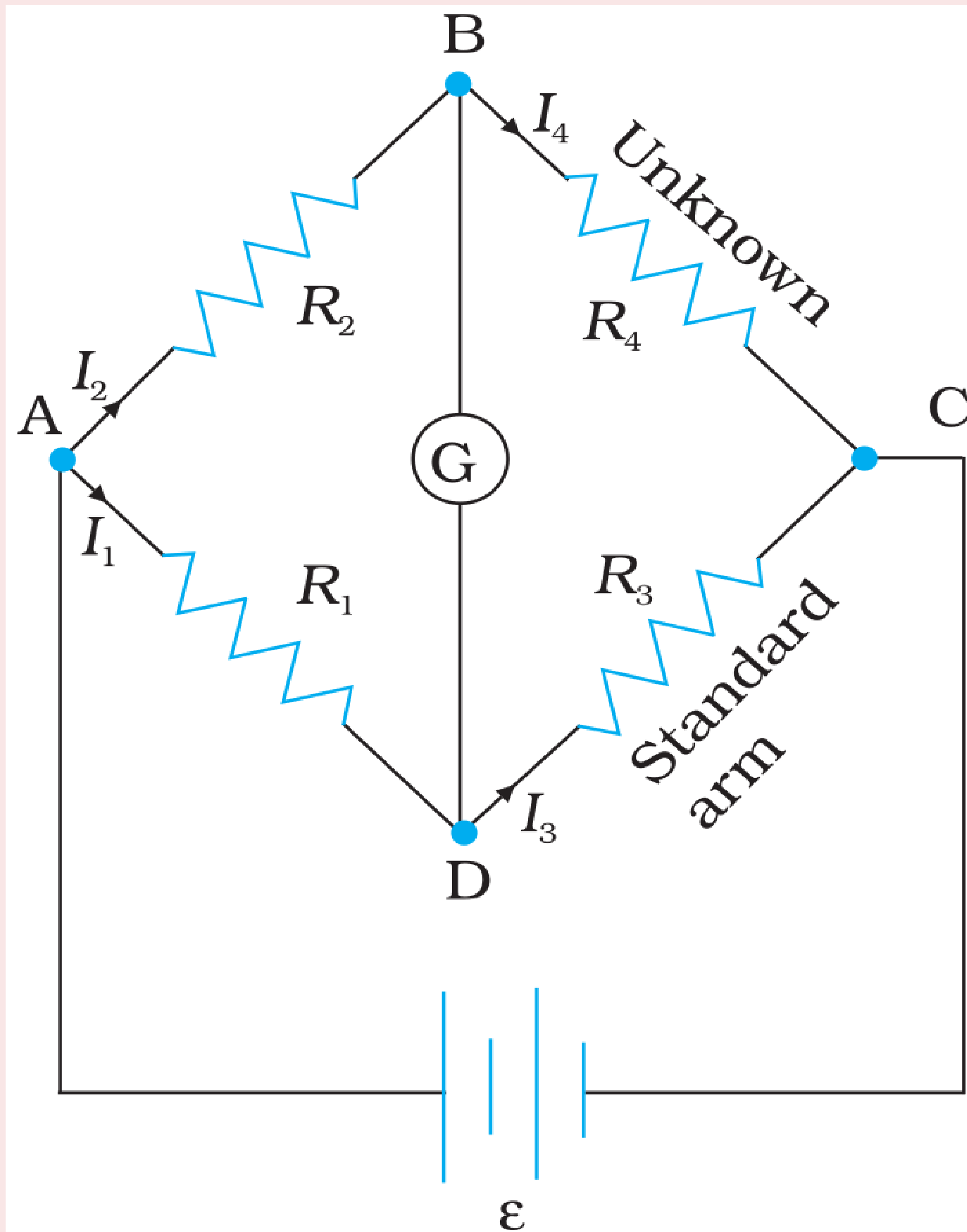
$$I_2 = I_4 \text{ and } I_1 = I_3$$

WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

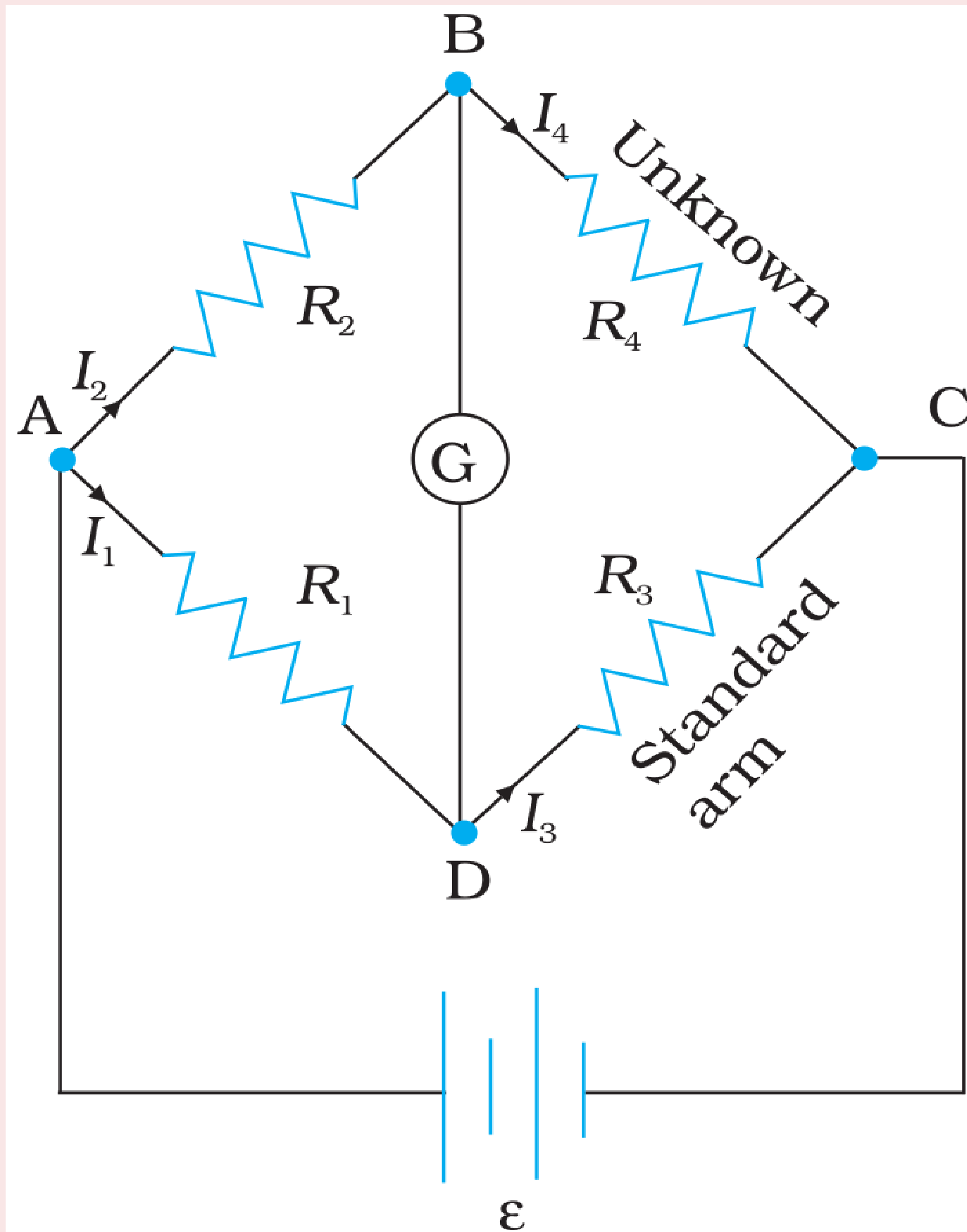
WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

$$I_2 R_2 - 0 + I_1 R_1 = 0 \quad \because (I_g = 0)$$

WHEATSTONE BRIDGE

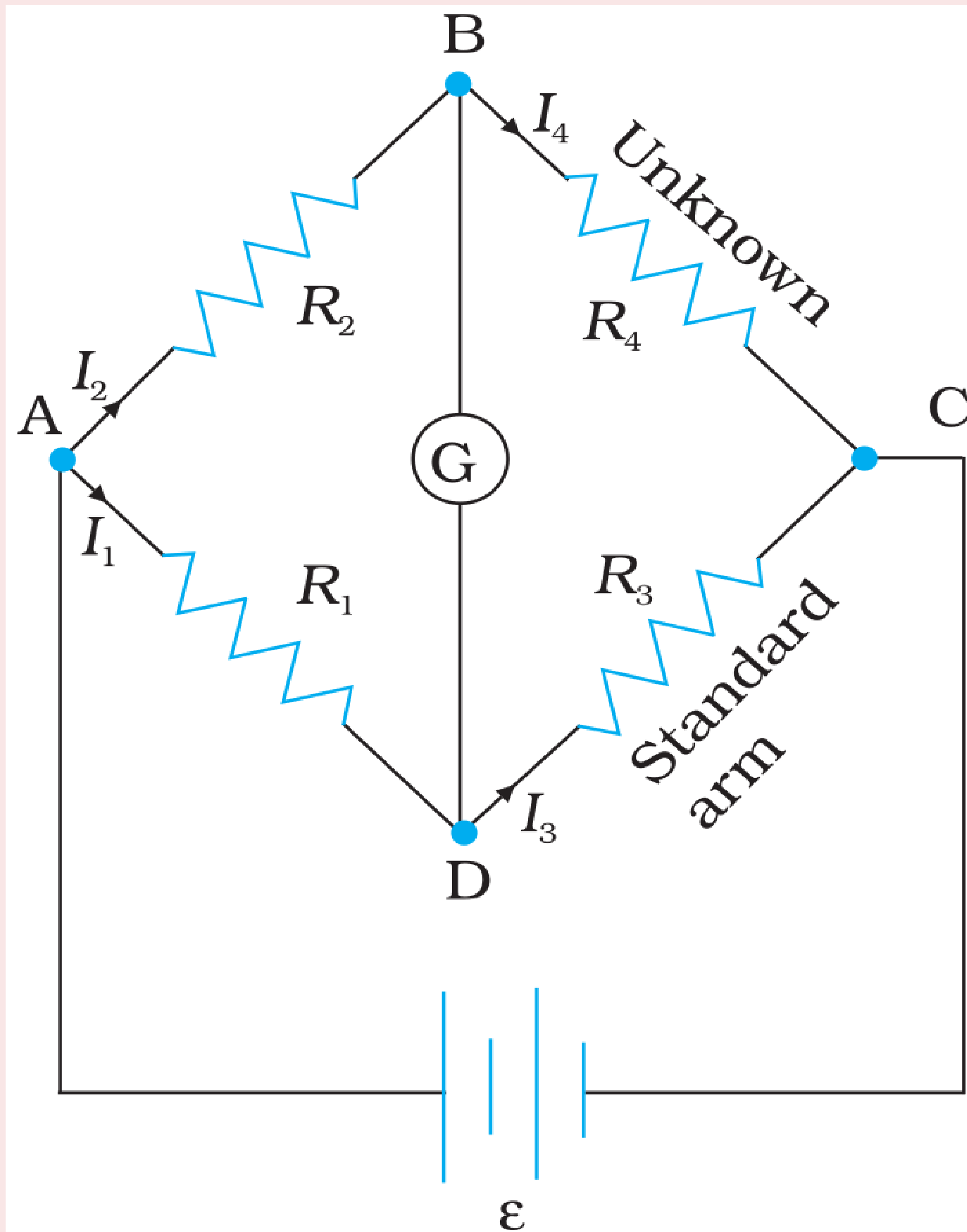


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WHEATSTONE BRIDGE



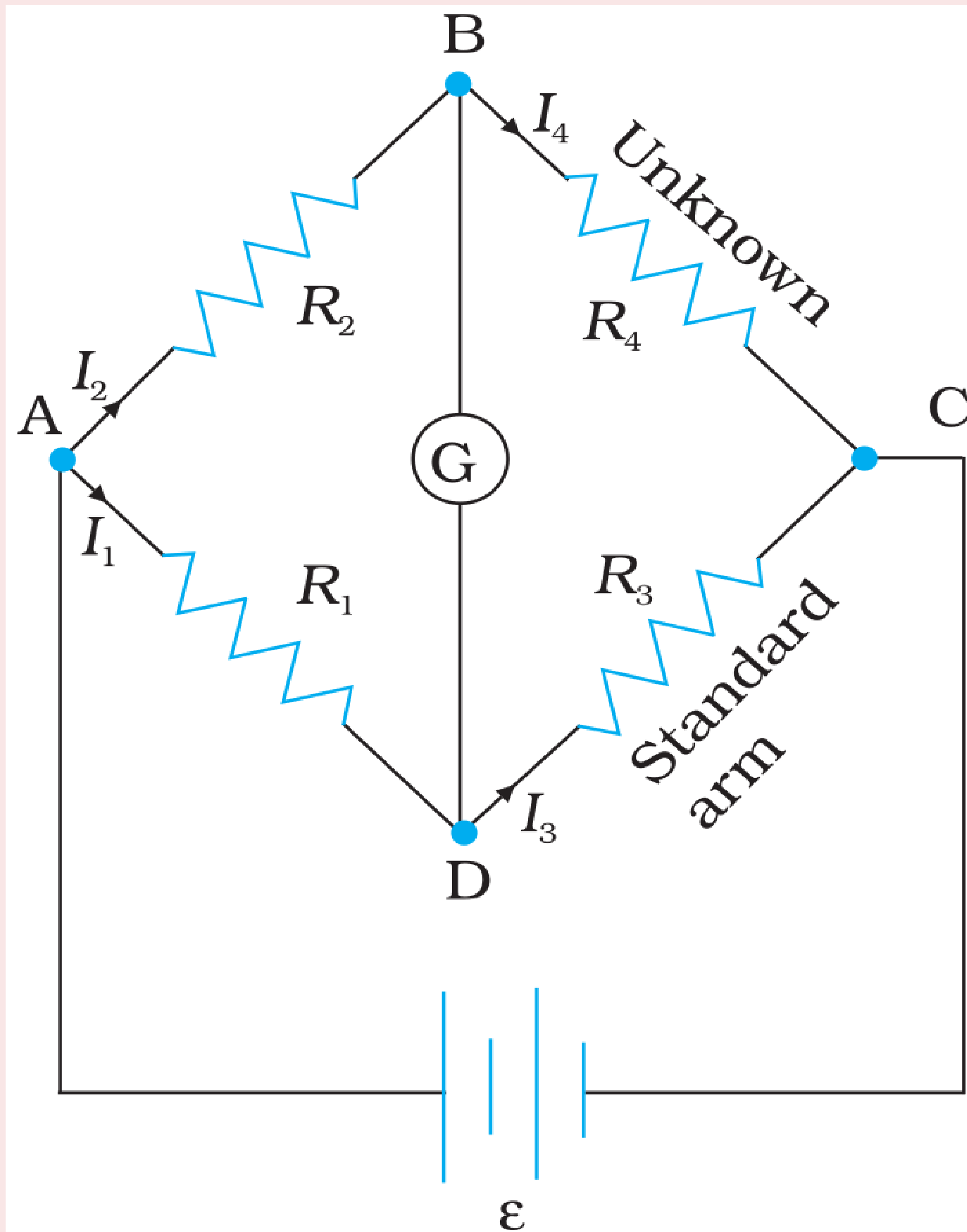
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WHEATSTONE BRIDGE



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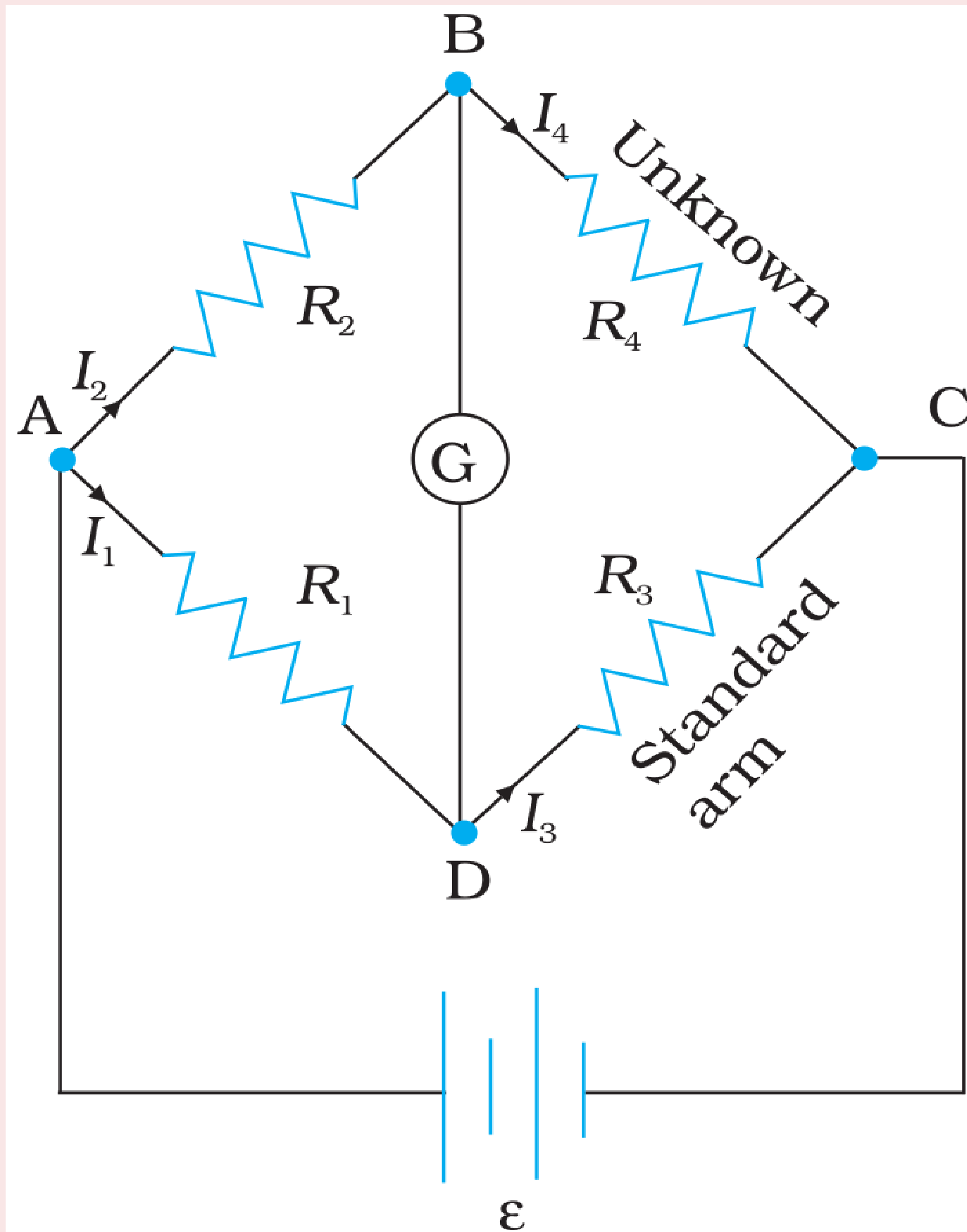
$$I_2 R_2 - 0 + I_1 R_1 = 0 \quad \because (I_g = 0)$$

$$\boxed{I_2 R_2 = I_1 R_1} \quad \dots\dots(1)$$

- Using Kirchhoff's **loop rule** to closed **loop BCDB**

$$I_4 R_4 - I_3 R_3 + 0 = 0 \quad \because (I_g = 0)$$

WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

$$I_2 R_2 - 0 + I_1 R_1 = 0 \quad \because (I_g = 0)$$

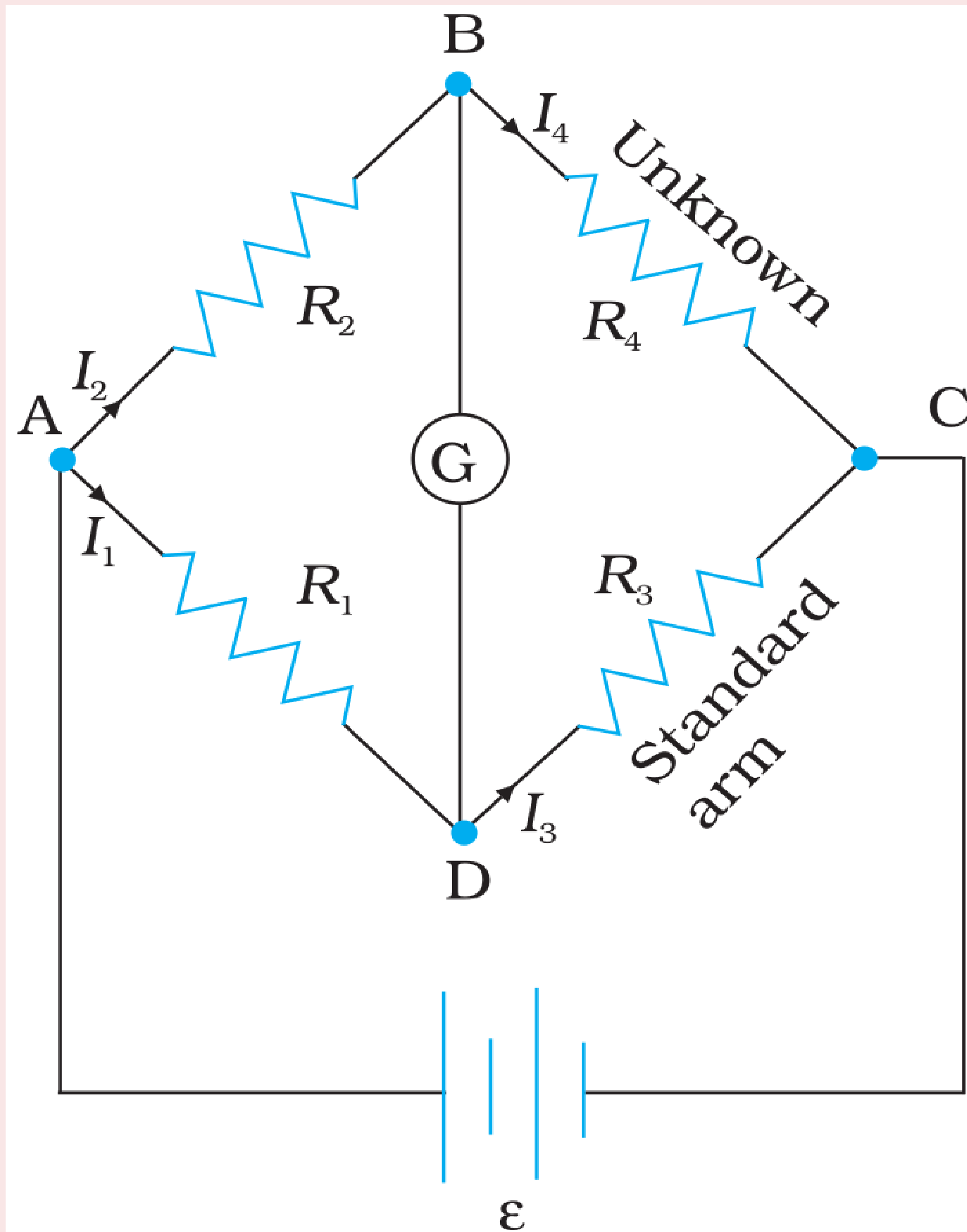
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- Using Kirchhoff's **loop rule** to closed **loop BCDB**

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$$\implies I_2 R_4 - I_1 R_3 = 0 \quad \because (I_4 = I_2 \text{ and } I_3 = I_1)$$

WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

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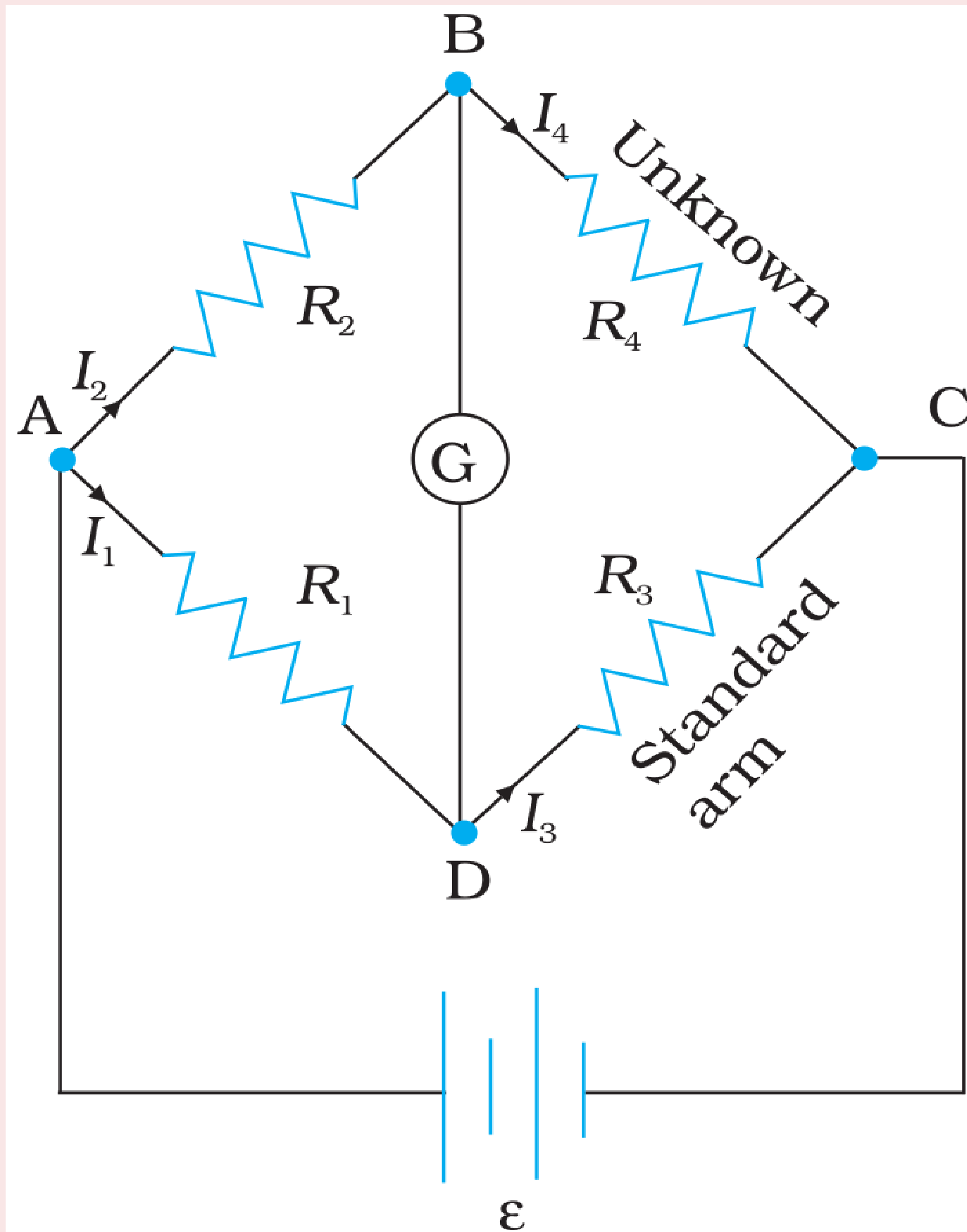
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WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

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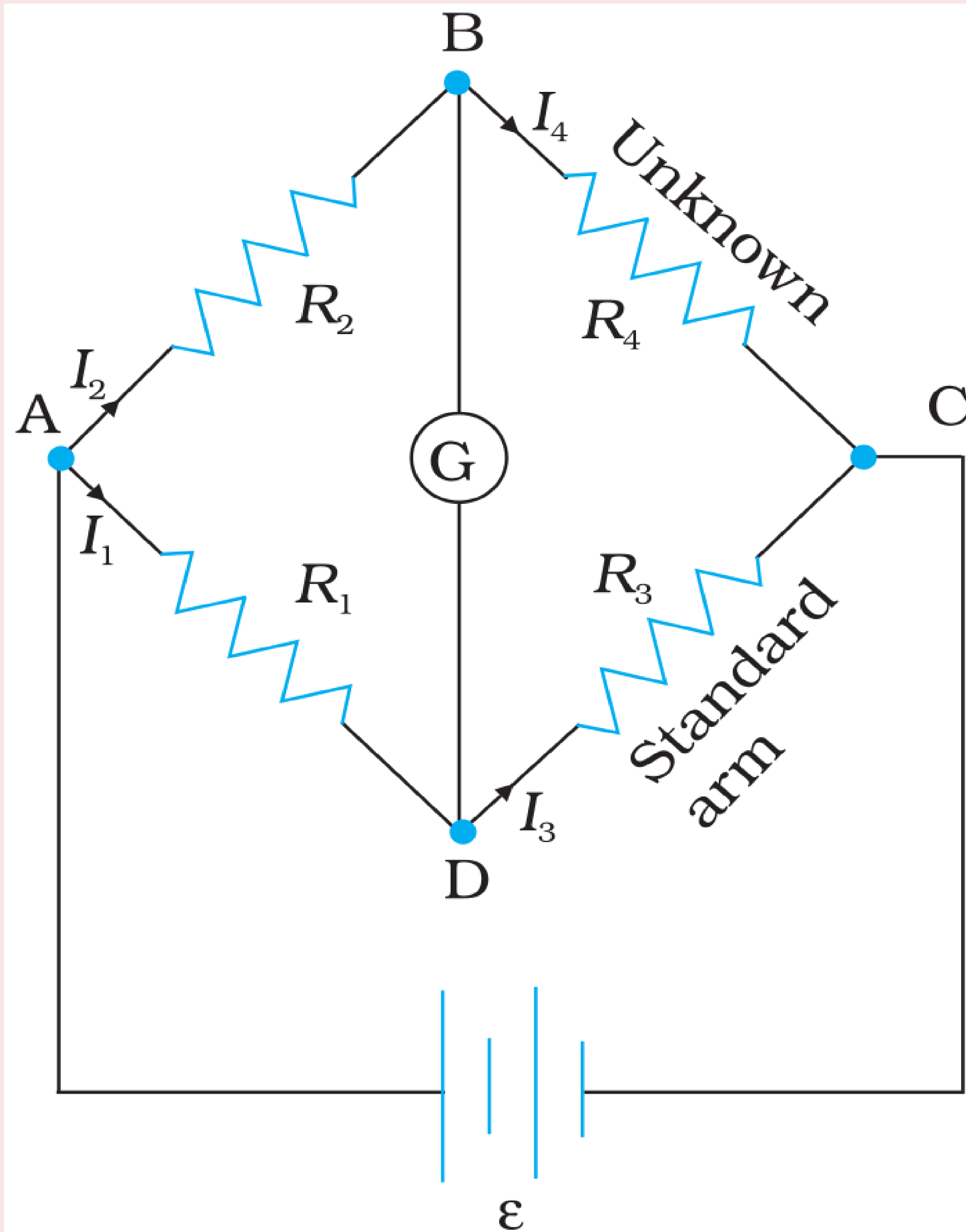
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$$\boxed{I_2 R_4 = I_1 R_3} \quad \dots\dots(2)$$

- Dividing eq(1) by eq(2)** we get:

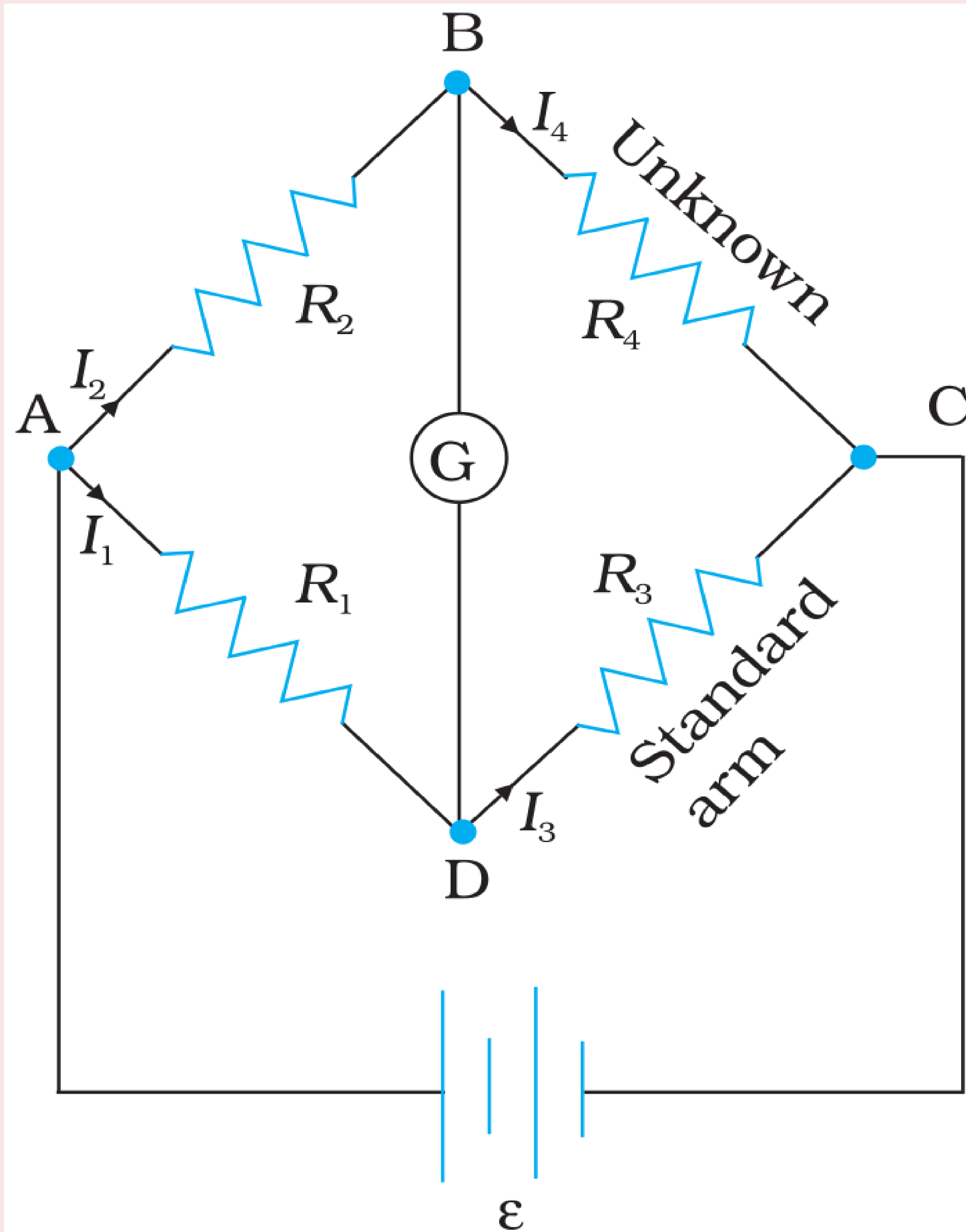
$$\boxed{\frac{R_2}{R_4} = \frac{R_1}{R_3}} \quad \dots\dots\dots(3)$$

USES OF WHEATSTONE BRIDGE



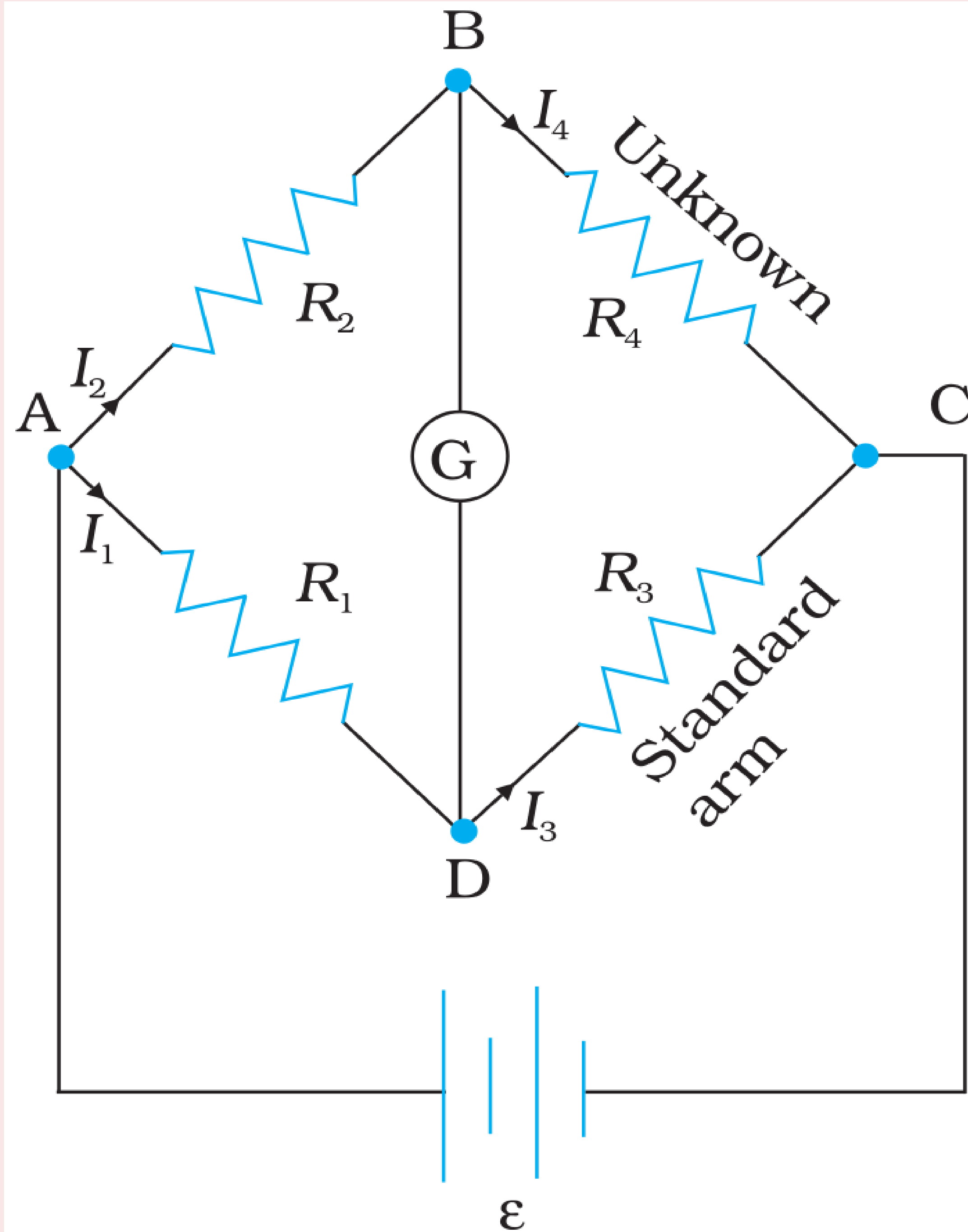
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USES OF WHEATSTONE BRIDGE



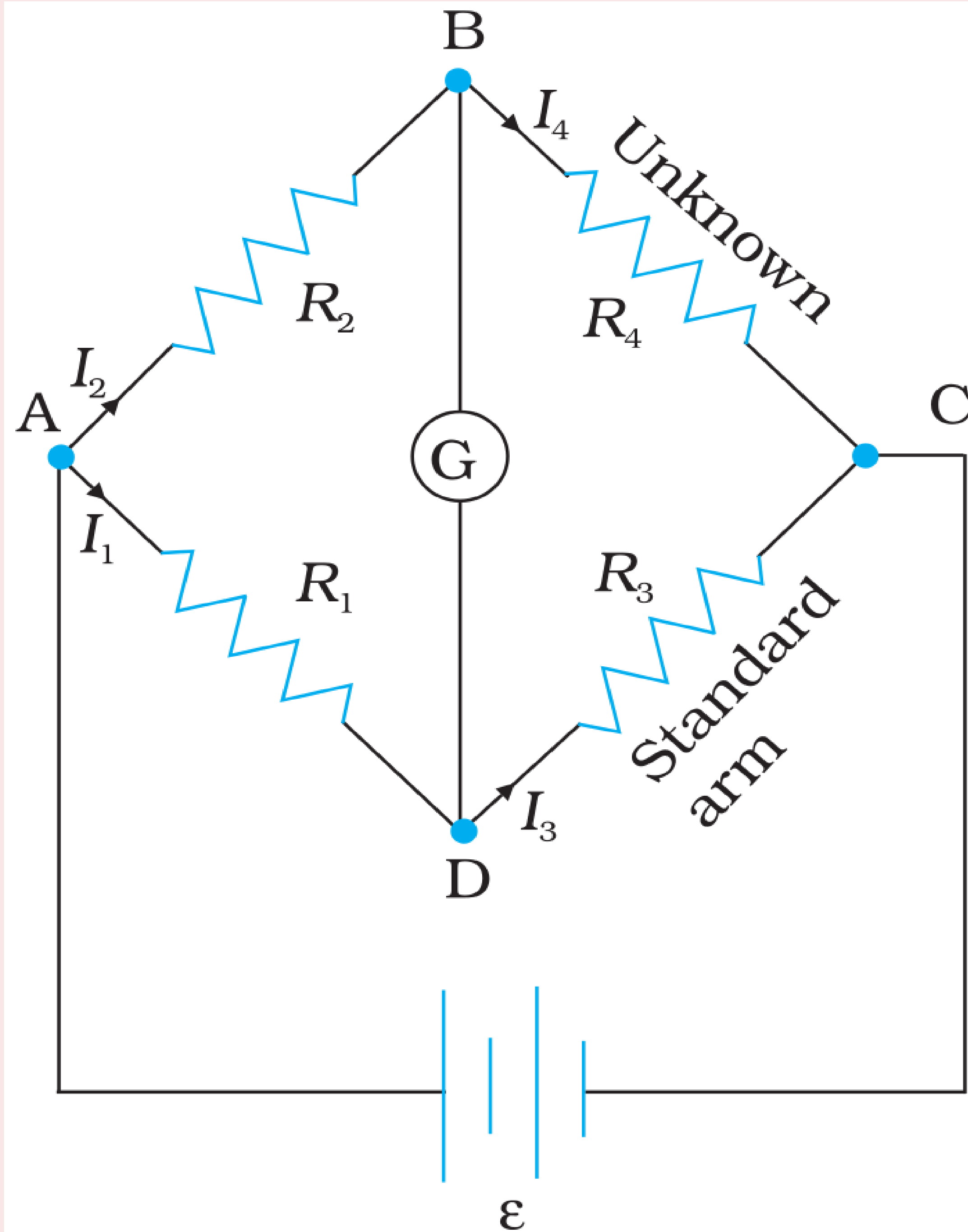
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USES OF WHEATSTONE BRIDGE



- The **Wheatstone bridge** and its **balance condition** provide a practical method for **determination** of an **unknown resistance**.
- Let, R_4 is the **unknown resistor**. and R_1 , R_2 are **known resistors**.
- We go on **varying R_3** till the **galvanometer** shows a **null deflection**. The bridge then is **balanced** and the **value of the unknown resistance R_4** is given by

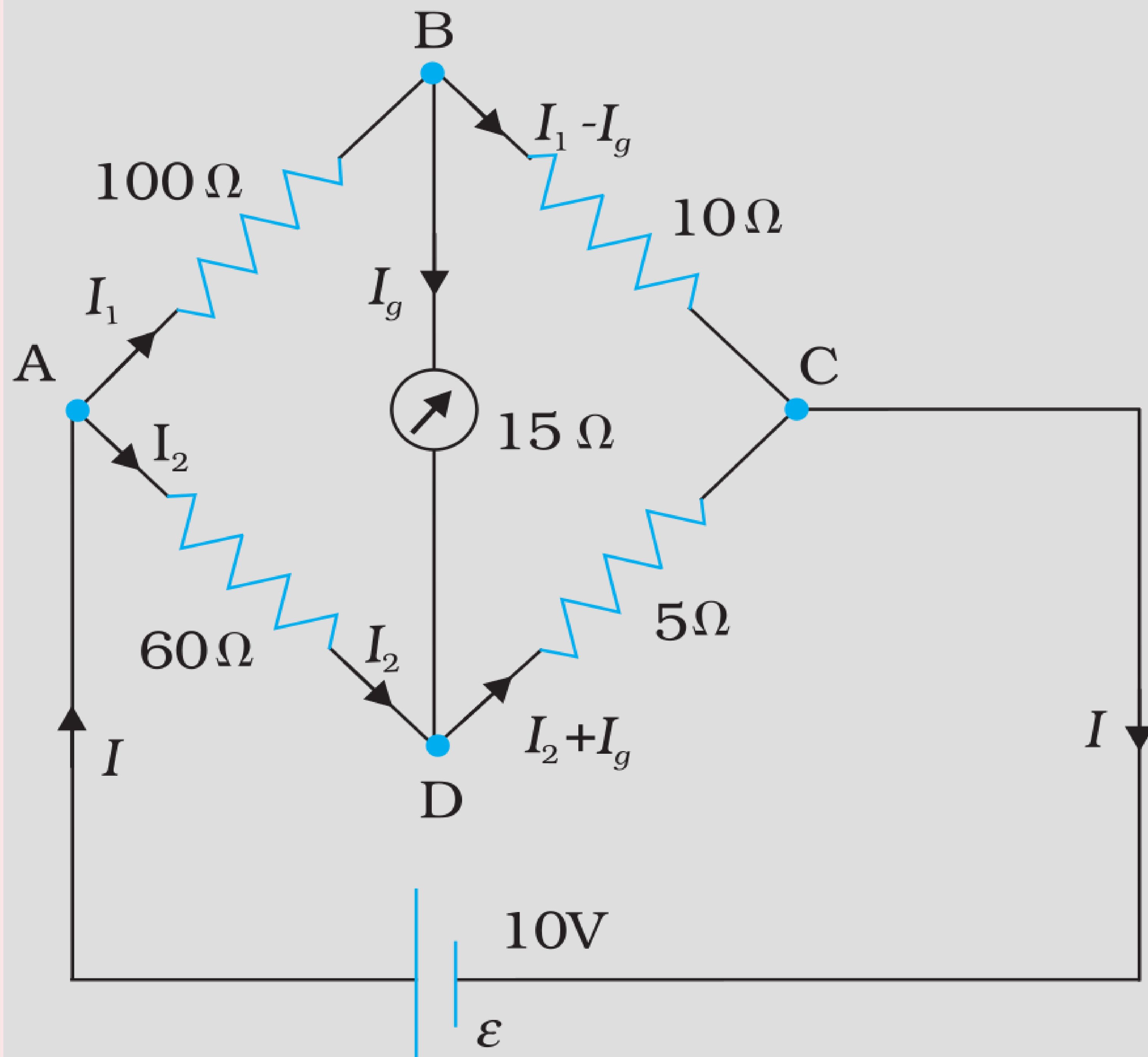
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$$R_4 = \frac{R_2 R_3}{R_1} \dots\dots(4)$$

WHEATSTONE BRIDGE



Example 3.8

The four arms of a Wheatstone bridge have the following resistances:

$AB = 100\ \Omega$, $BC = 10\ \Omega$, $CD = 5\ \Omega$, and $DA = 60\ \Omega$.